

Principles of bionics and biorobotics engineering

Bioelectronics and neuroengineering in
neurology and neurorehabilitation – Part 2

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Outline

1. Background on nervous system injuries and disorders
- 2. Technologies for neurorehabilitation**
3. Technologies for functional assessment

2. Technologies for neurorehabilitation

2.1 Stimulation-based interventions

- Brain stimulations
- Spinal cord stimulations
- Functional electrical stimulation

2.2 Brain-computer interfaces (BCIs)

- Overview of BCIs
- BCI paradigms
- Types of BCIs and applications

2.3 Neural interface technologies

- Framework and components of a neuroprosthesis
- Interfaces with the CNS and PNS

2. Technologies for neurorehabilitation

2.1 Stimulation-based interventions

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Neurostimulation for treating neurological disorders

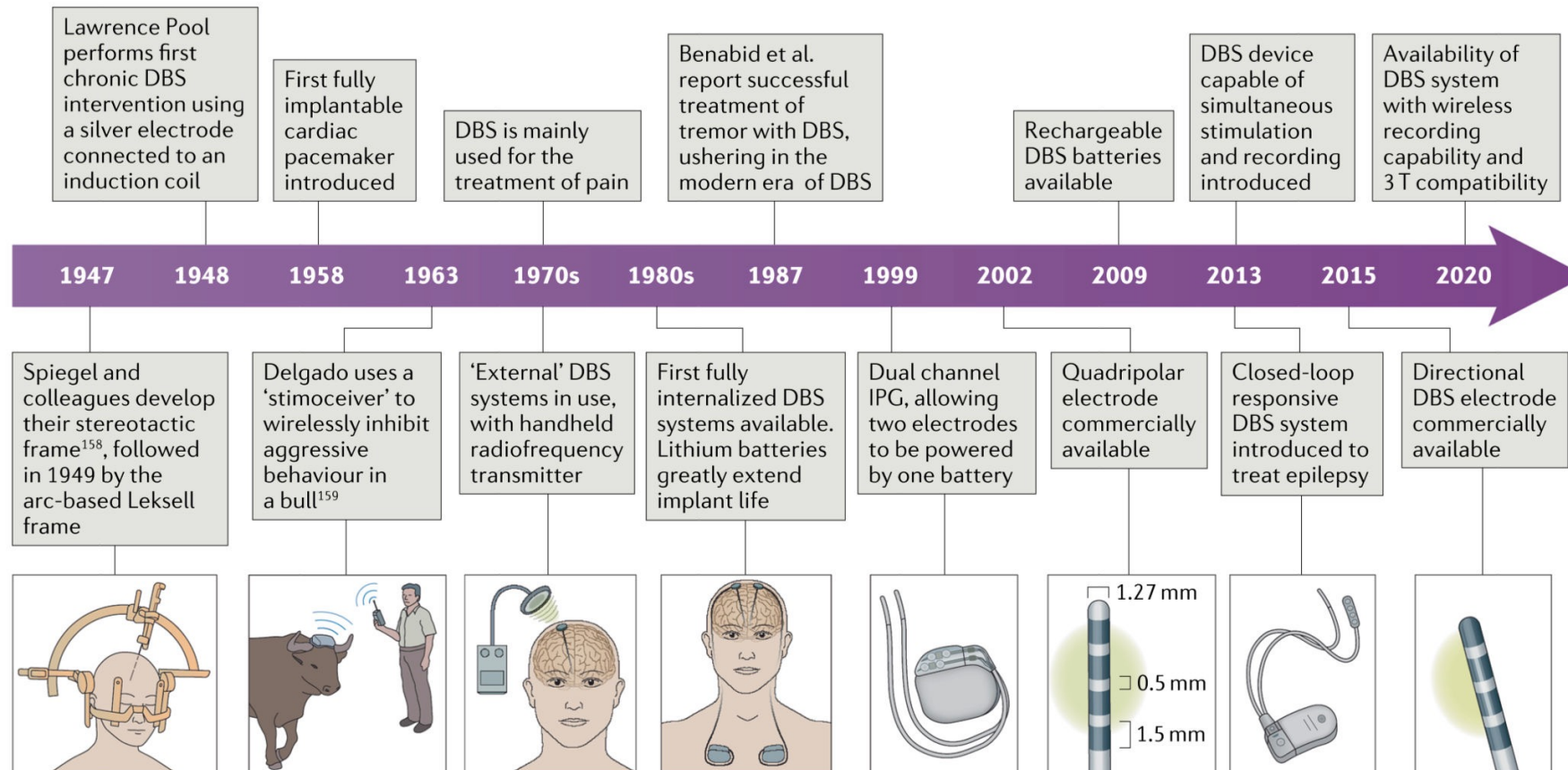
- Advances in neurostimulation are creating new treatment options for patients with severe neurological and psychiatric conditions.
- Neurostimulation therapies include both **invasive** and **non-invasive** approaches that apply targeted electromagnetic energy to modulate specific neural circuits.
- Neural modulation is the process of adjusting neural activity to alter the function of brain or nervous system circuits.
- This can either enhance or inhibit electrical signaling within targeted areas, leading to therapeutic effects and improved neurological functions.
- Methods include electrical, magnetic, and chemical stimulation, with applications in treating chronic pain, movement disorders, epilepsy, and psychiatric conditions.

Neurostimulation for treating neurological disorders

- **Brain** stimulations
 - Deep brain stimulation (DBS)
 - Non-invasive brain stimulation (TDCS and TMS)
- **Spinal cord** stimulations
 - Epidural electrical stimulation (EES)
 - Transcutaneous spinal cord stimulation (tSCS)
- **Functional electrical** stimulation (FES) and therapy (FET)

Deep brain stimulation (DBS)

DBS is a neurosurgical procedure that enables precise, circuit-based neuromodulation. It is a standard treatment for Parkinson's disease, essential tremor, and dystonia. DBS is also being actively researched for other conditions involving disrupted neural circuits, such as major depressive disorder and Alzheimer's disease.

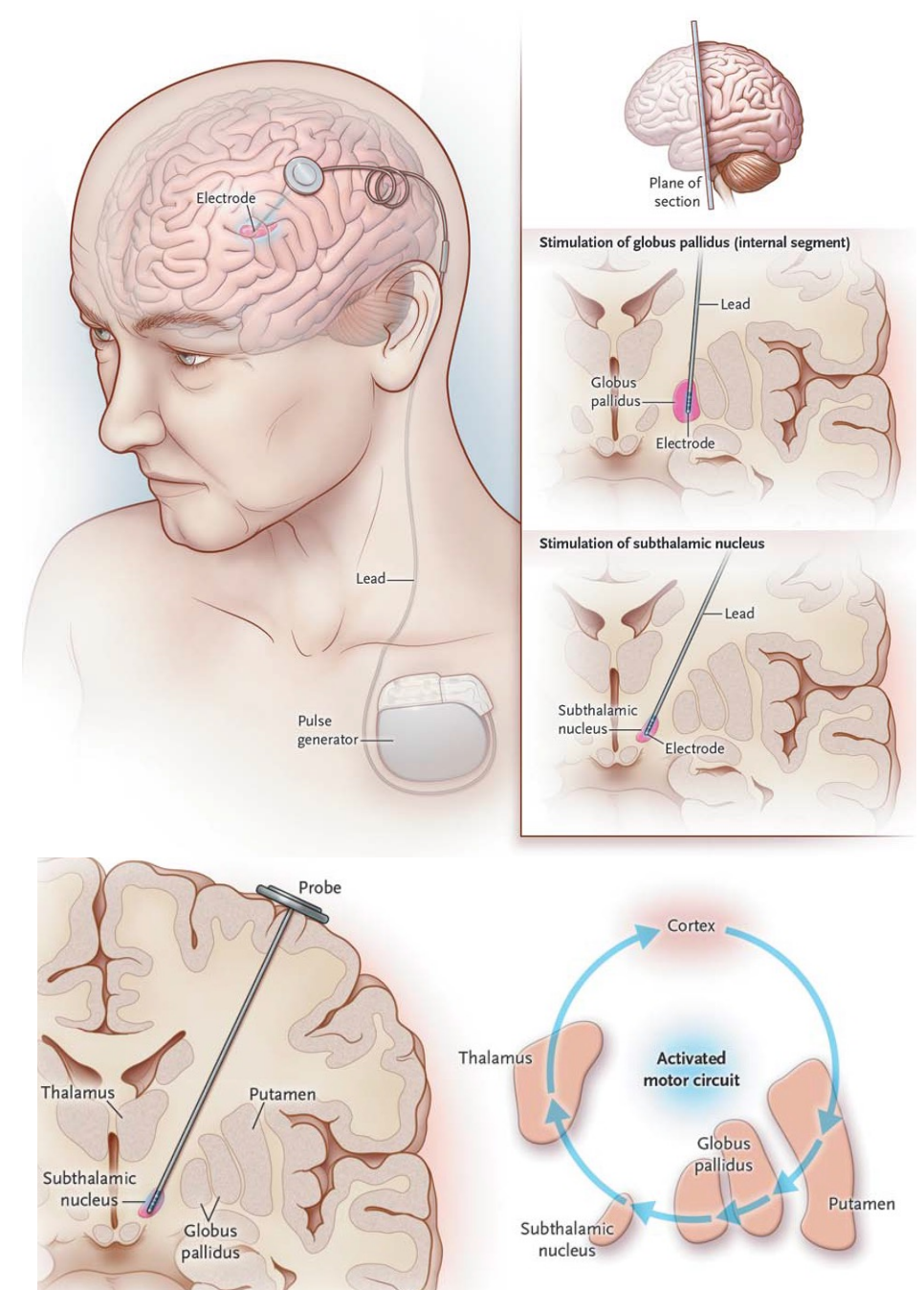


Deep brain stimulation (DBS)

- DBS represents the most significant therapeutic advancement for PD since the introduction of levodopa. It is particularly effective for patients with severe tremors and medication-induced dyskinesias, with research showing benefits that can last at least five years.
- **DBS is not suitable for everyone with PD**, as certain symptoms (e.g., speech difficulties, swallowing issues, cognitive changes, and gait freezing) do not reliably improve with DBS therapy.
- In DBS surgery, electrodes are implanted into a specific area of the brain, guided by MRI and, occasionally, real-time recordings of brain cell activity during the operation.
- A second procedure is conducted to place an implantable pulse generator (IPG), similar to a heart pacemaker and is roughly the size of a stopwatch.
- The IPG is situated under the collarbone or in the abdomen, where it provides electrical stimulation to brain areas responsible for movement. Patients who receive DBS surgery are provided with a controller that allows them to turn the device on or off and monitor basic settings like battery life.

Deep brain stimulation (DBS)

- Electrodes are implanted in the brain and connected to an IPG that delivers electrical stimuli to modulate neural signaling.
- The most common targets for DBS in PD are the **subthalamic nucleus** and the **internal globus pallidus**, both located in the basal ganglia.
- DBS alters the firing rates and patterns of neurons, affects synaptic activity, and triggers nearby astrocytes to release neurotransmitters, resulting in increased blood flow and neurogenesis.
- The cumulative effects of DBS extend beyond local neurons, influencing larger neural networks, although the exact mechanisms by which it alleviates PD symptoms remain unclear.



Deep brain stimulation (DBS)

IPG

- It is the core component of DBS. It houses a battery and power module, a CPU with program memory, and a microprocessor that oversees all device functions, including activation, deactivation, pulse parameters, internal diagnostics, and communication with external devices.
- Some IPGs feature recharging capabilities, integrated accelerometers, local field potential (LFP) sensing, as well as onboard signal processing and analysis functionalities.



Model	St. Jude (Abbott) Infinity 5	St. Jude (Abbott) Infinity 7	Boston Scientific Vercise PC	Boston Scientific Vercise Gevia	Boston Scientific Vercise Genus P8/P16
No. of channels	2	2	2	2	1/2
Weight (gram)	49	58	55	26	58
Size (mm)	56×50×13	67×50×14	71×50×11	51×46×11	72×50×12
RC	No	No	No	Yes	No
Freq. range (Hz)	2–240	2–240	2–255	2–255	2–255
Pulse width (μs)	20–500	20–500	10–450	20–450	20–450
TF	MSS	MSS	A	A	A
CF	CA	CA	MICC	MICC	MICC
Directional lead	Yes	Yes	Yes	Yes	Yes
MRI safety	C	C	U	C	C
LFP sensing	No	No	No	No	No



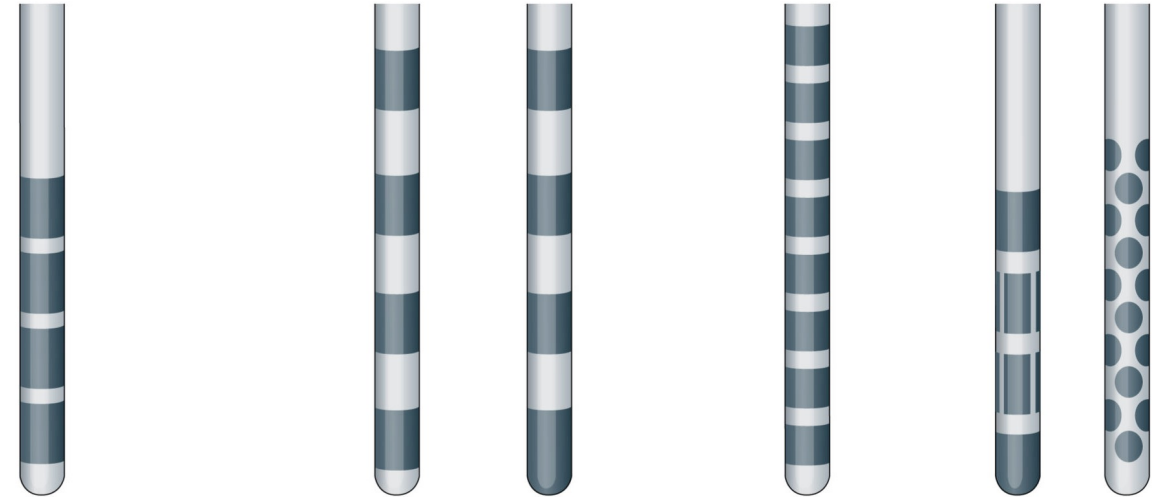
Model	Boston Scientific Vercise Genus R16	Medtronic Activa PC	Medtronic Activa RC	Medtronic Activa SC	Medtronic Percept PC
No. of channels	2	2	2	1	2
Weight (gram)	27	67	40	44	61
Size (mm)	52×46×11	65×49×15	54×54×9	55×60×11	68×51×12
RC	Yes	No	Yes	No	No
Freq. range (Hz)	2–255	2–250	2–250	3–250	2–250
Pulse width (μs)	20–450	60–450	60–450	60–450	20–450
TF	A	IL	IL	IL	IL
CF	MICC	No	No	No	No*
Directional lead	Yes	No	No	No	No
MRI safety	C	C	C	C	C
LFP sensing	No	No	No	No	Yes

A, areas; **C**, conditional; **CA**, coactivation; **CF**, current fractionation; **IL**, interleaving; **LFP**, local field potential; **MICC**, multiple independent current control; **MSS**, multi-stim set; **PC**, primary cell; **RC**, rechargeable cell; **SC**, single cell; **TF**, temporal fractionation; **U**, unsafe

Deep brain stimulation (DBS)

Electrodes

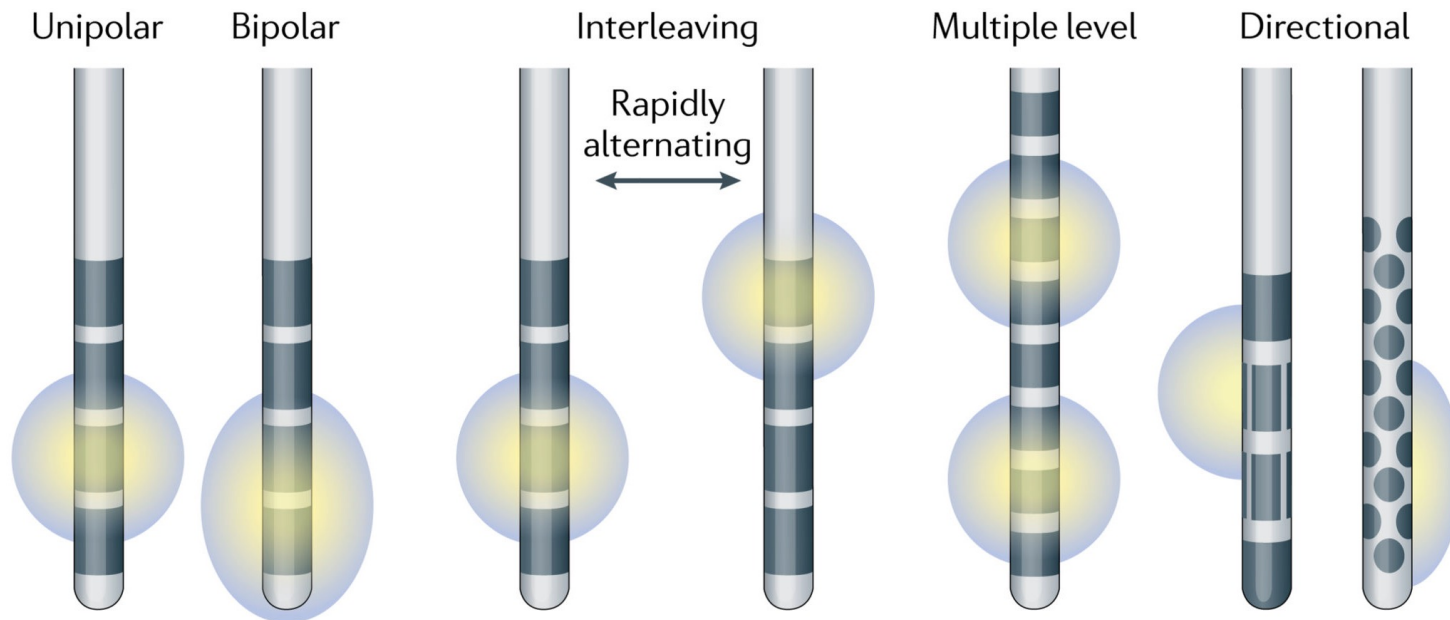
- Electrodes must be biocompatible, durable, stable, and have good conductivity, while also considering surgical feasibility, MRI compatibility, and sensing capabilities.
- Platinum-iridium wires and nickel alloy connectors. DBS electrodes are encased in a polyurethane sheath for low toxicity and high conductivity, typically featuring four stimulating contacts.
- Greater spacing between contacts increases the range of neural targets, while closer spacing allows for more precise stimulation control.
- Directional electrodes enable more flexible shaping of the electric field, enhancing therapeutic outcomes by improving efficacy and minimizing side effects; they utilize radially segmented contacts for horizontal movement and current direction.
- While offering advanced capabilities due to increased contact options, directional electrodes complicate surgical implantation and pose challenges in programming.



Deep brain stimulation (DBS)

Modes of Stimulation

- **Unipolar:** current directed from the battery to a single contact (or vice versa).
- **Bipolar:** current flows between two contacts, with at least one acting as an anode and the other as a cathode.
- **Interleaving:** alternates between different stimulation settings to optimize therapy.
- **Multiple level:** stimulation of multiple neural targets along the electrode trajectory.
- **Directional:** current is directed or shaped based on local anatomy or specific clinical symptoms, enhancing therapeutic precision.



Deep brain stimulation (DBS)

Modes of Stimulation

Conventional asymmetric biphasic waveform featuring a brief cathodic phase followed by an interphase delay and a prolonged anodic (recharge) phase

a

Symmetric biphasic waveform with equal-duration cathodic and anodic phases

b

Symmetric biphasic waveform with no interphase delay

c

d

Reversal of the standard pulse phase order in a symmetric biphasic waveform

e

f

g

h

Regular temporal stimulation pattern with fixed interpulse intervals (approximately 7.7 ms or 130 Hz)

Non-regular temporal stimulation pattern characterized by random interpulse intervals.

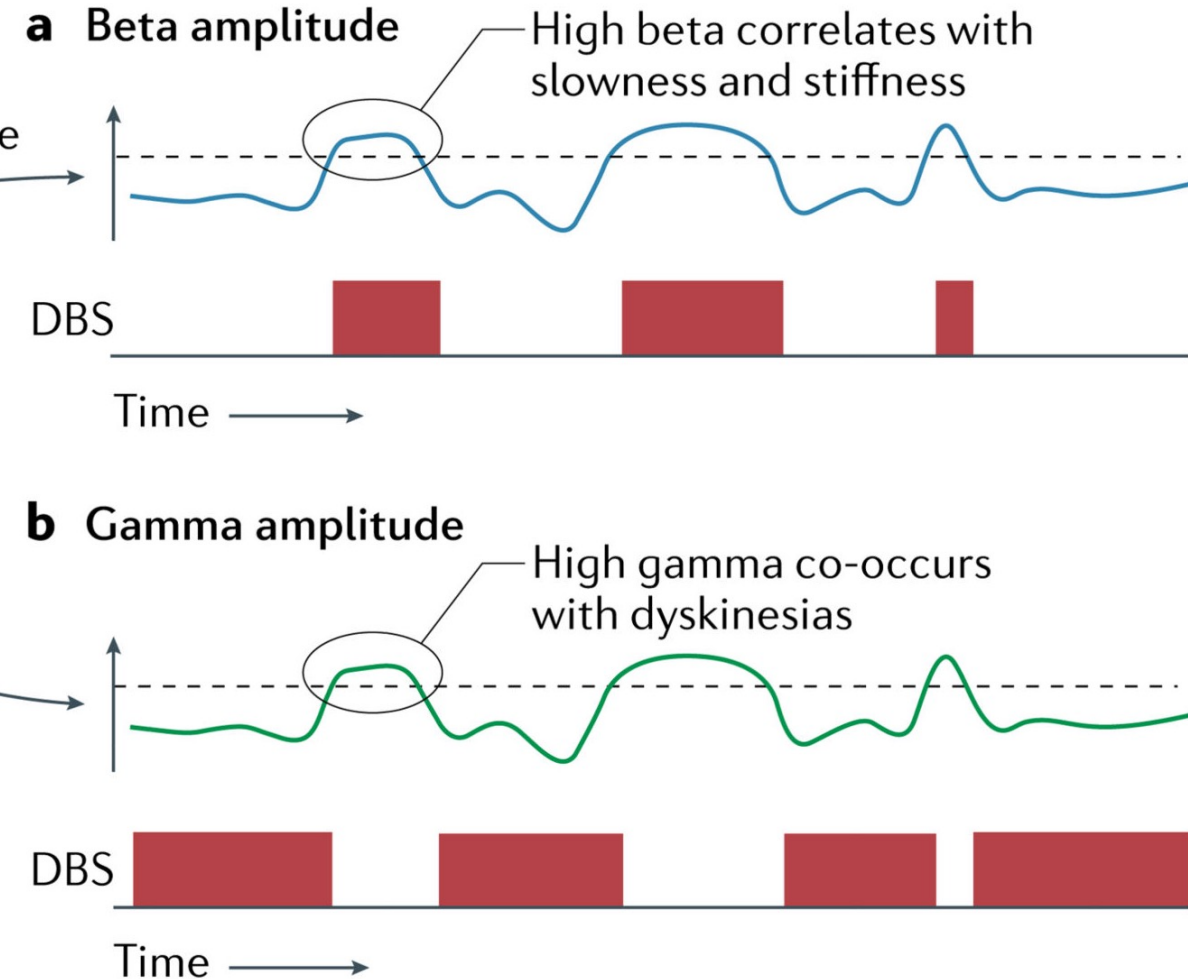
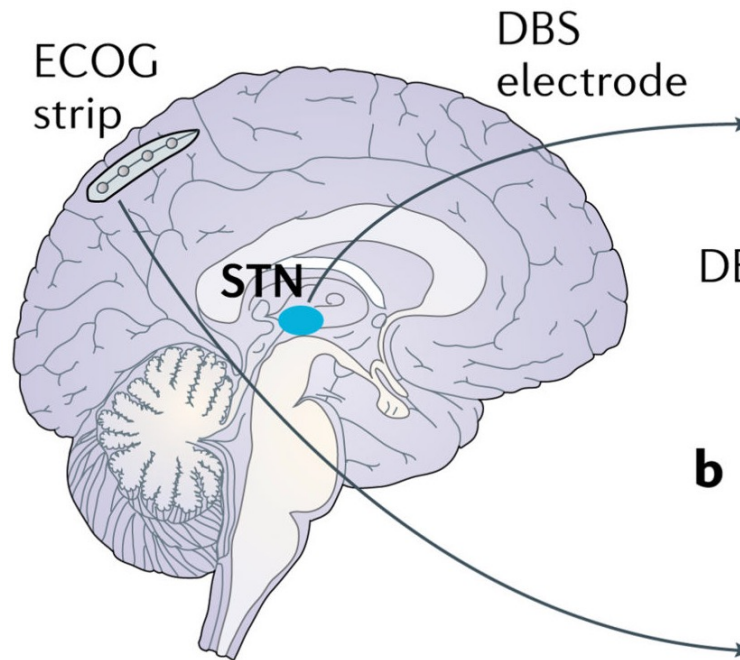
Burst stimulation pattern comprising several pulses at short interpulse intervals followed by a longer interval.

Coordinated reset stimulation pattern involving bursts across four different electrode contacts, with each row representing the pattern for each contact.

Deep brain stimulation (DBS)

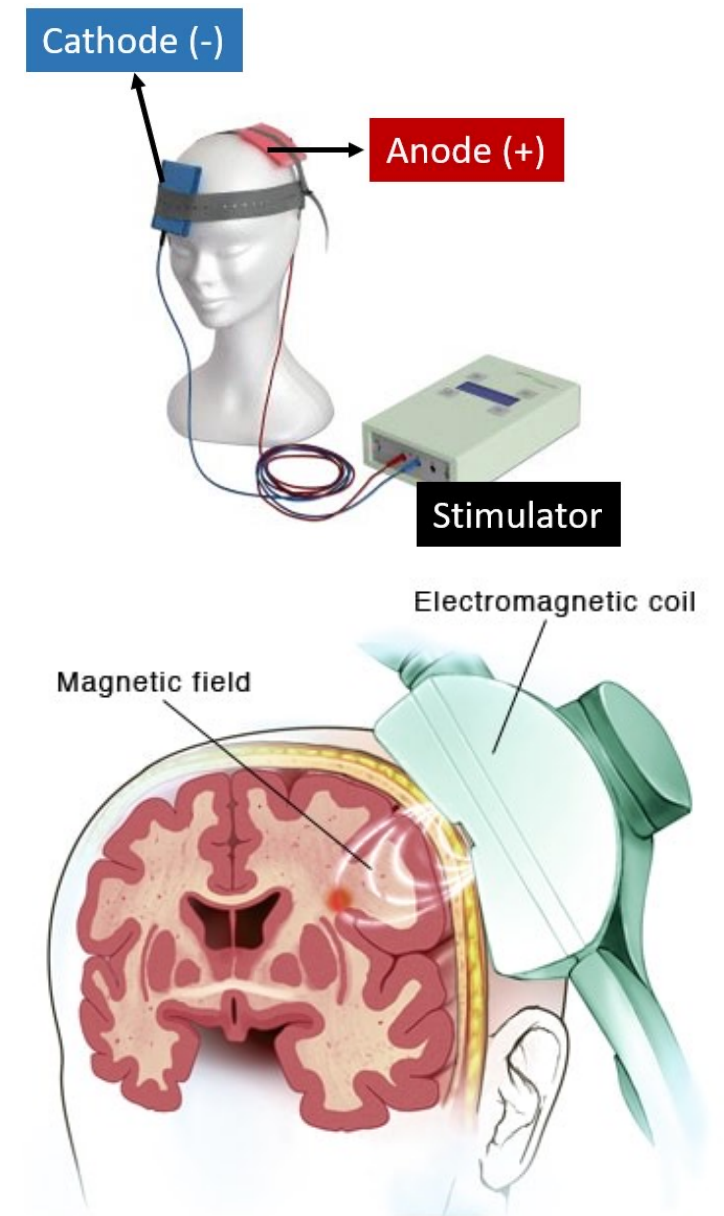
Adaptive DBS (aDBS)

- The ability to adjust DBS based on one or more feedback signals over time.
- aDBS for PD primarily uses LFP feedback in the beta frequency range to monitor bradykinesia and rigidity, while gamma activity from cortical electrodes serves as an indicator of dyskinesia.
- This feedback mechanism allows for more efficient and responsive DBS treatment in managing motor symptoms of PD.



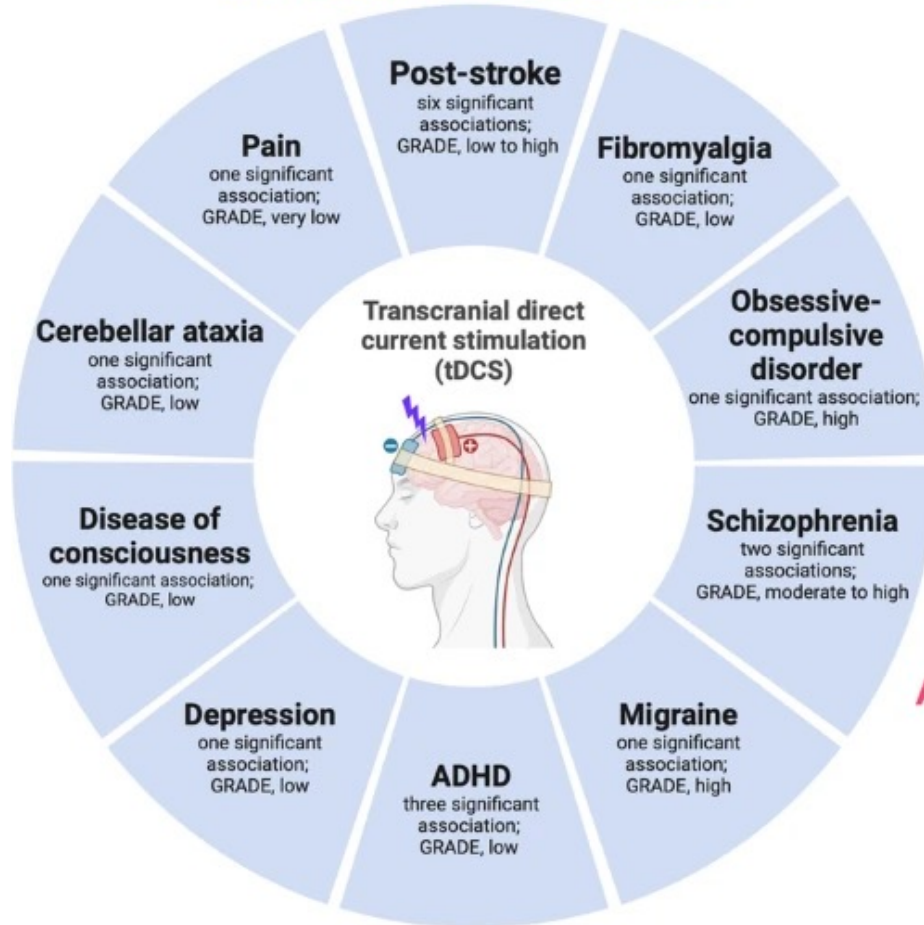
Non-invasive brain stimulations

- Non-invasive brain stimulation (NIBS) is increasingly used in rehabilitation for conditions such as stroke, spinal cord injury, traumatic brain injury, and multiple sclerosis, as well as for diagnostic neurophysiological assessments.
- The two primary NIBS techniques are transcranial magnetic stimulation (TMS) and transcranial direct current stimulation (tDCS)
- When used as an adjunct to traditional rehabilitation, NIBS aims to modulate neural activity in targeted cortical areas, either by inhibiting or activating specific brain regions.
- Therapeutic NIBS is indicated for various neurorehabilitation purposes, including motor recovery, aphasia, neglect, dysphagia, cognitive disorders, spasticity, and central pain.



Transcranial direct current stimulation (TDCS)

Therapeutic effects of tDCS



Adverse effect of tDCS

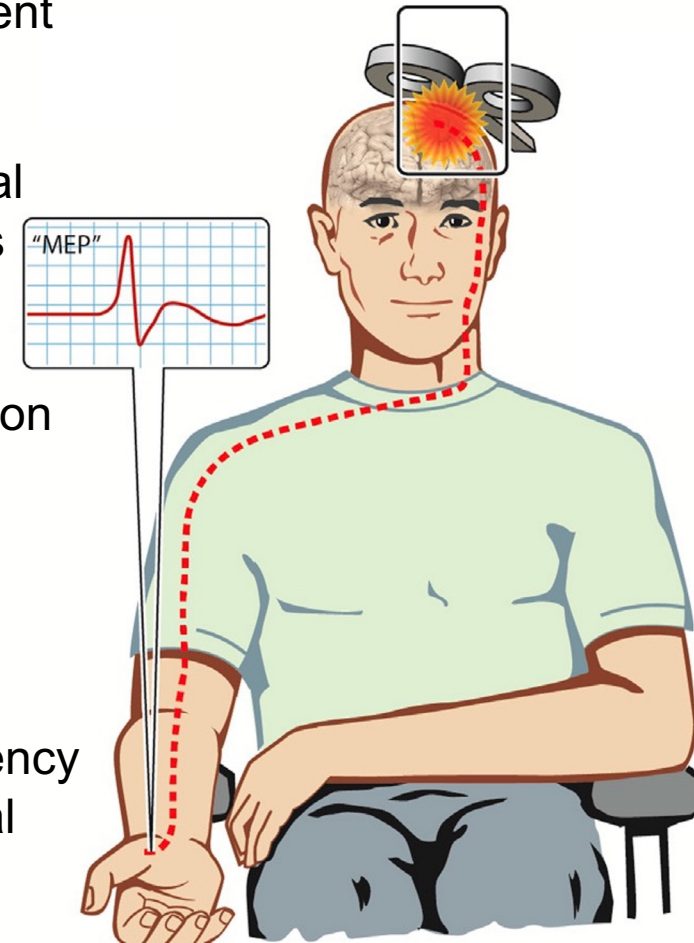
Treatment-emergent mania or hypomania
one significant association;
GRADE, moderate

- TDCS is a non-invasive brain stimulation technique that involves applying a low electrical current to the scalp using electrodes, to modulate brain activity by altering neuronal excitability
- Current flows through the brain, affecting the cortical regions beneath the electrodes and can increase (anodal stimulation) or decrease (cathodal stimulation) neural activity,

Abbreviations: ADHD, attention-deficit/hyperactivity disorder; GRADE, Grading of Recommendations, Assessment, Development, and Evaluation ; tDCS, transcranial direct current stimulation

Transcranial magnetic stimulation (TMS)

- TMS is a non-invasive brain stimulation technique that involves passing a current through a coil to generate a time-varying electromagnetic field over the scalp.
- The magnetic field induces an electric field in the brain tissue, activating cortical output cells, especially in the primary motor cortex (M1). This activation triggers corticospinal volleys, reflected as motor evoked potentials (MEPs).
- TMS effects depend on coil type (focal or non-focal), pulse waveform, stimulation strength, pulse number, and current direction, each activating different neural elements.
- Applications in **major depression** – treatment for depression when other therapies are ineffective; **obsessive-compulsive disorder (OCD)** – FDA-approved treatment for OCD; **migraine** – helps manage and reduce the frequency of migraines; **smoking cessation** – assists in quitting smoking when traditional methods fail; **post-stroke rehabilitation** – aids motor recovery after a stroke.



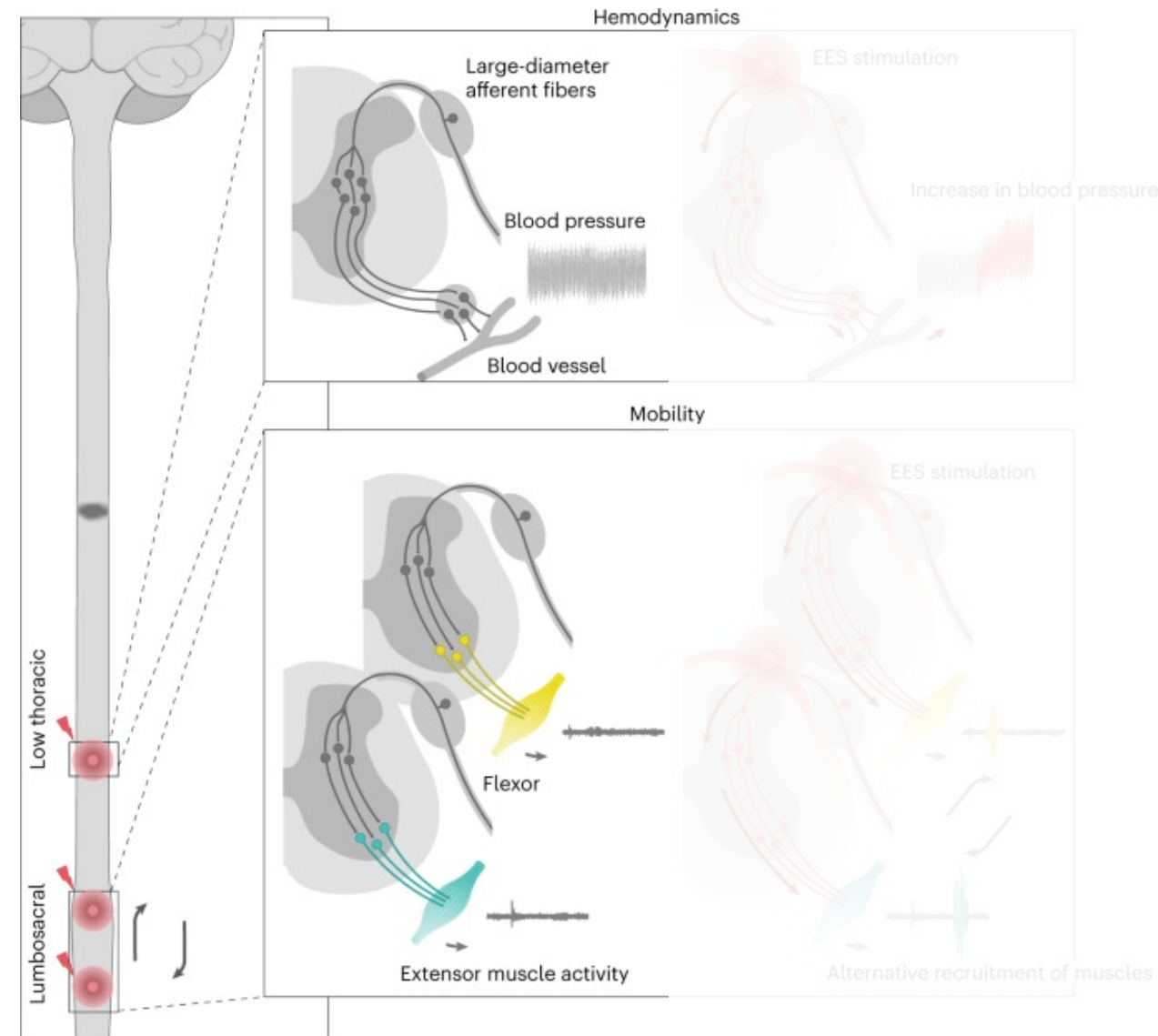
Epidural electrical stimulation (EES)

The sensorimotor circuits that produce muscle synergies are embedded in the spinal cord.

Descending signals adjust the responsiveness of these feedback circuits to match task-specific demands.

In spinal cord injury (SCI):

- Communication between the spinal cord and brain centers is disrupted.
- Descending pathways are interrupted, altering the task-specific modulation of sensorimotor feedback.
- The sensorimotor circuits lose essential sources of excitation and modulation needed to produce movement.

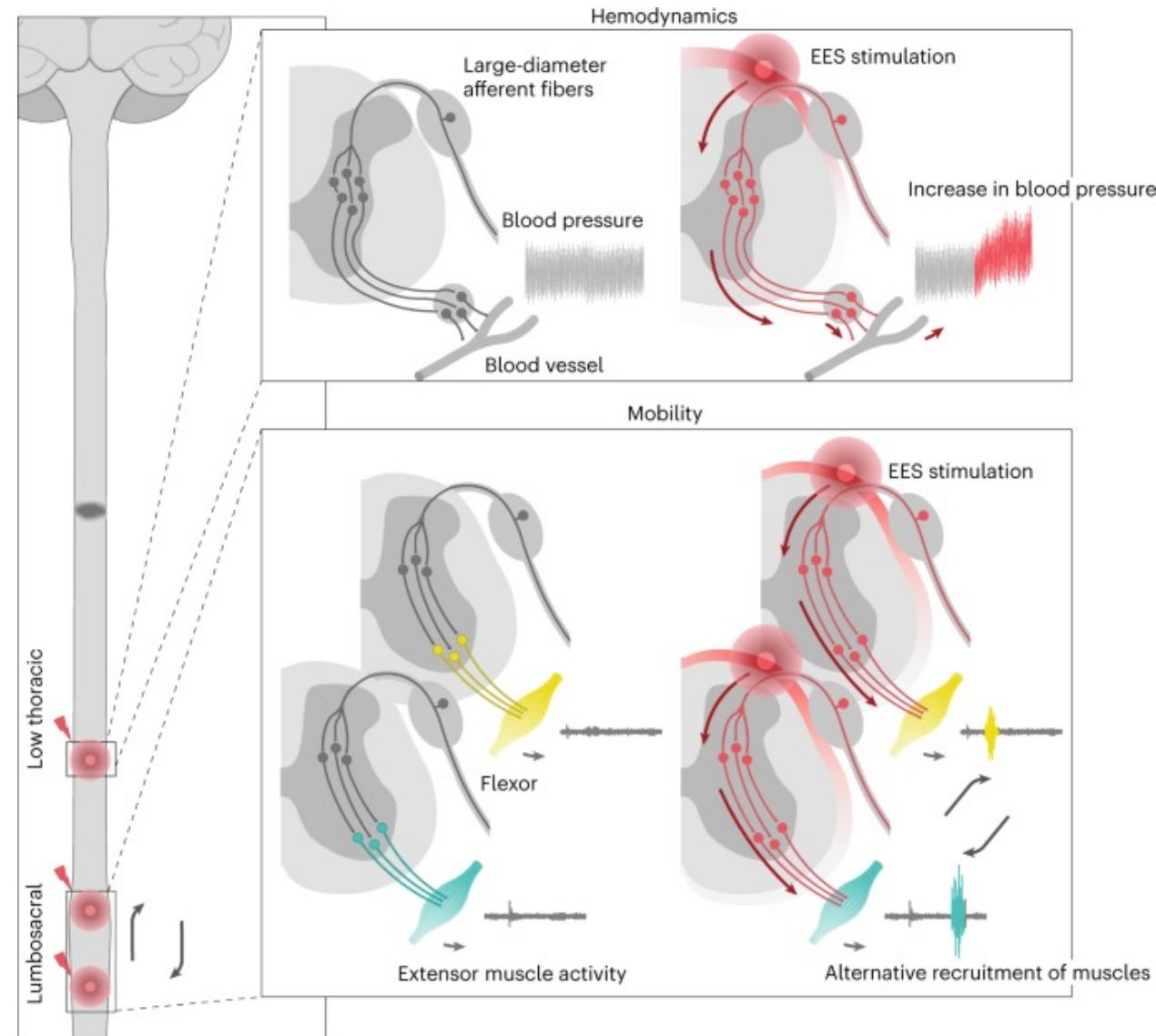


Epidural electrical stimulation (EES)

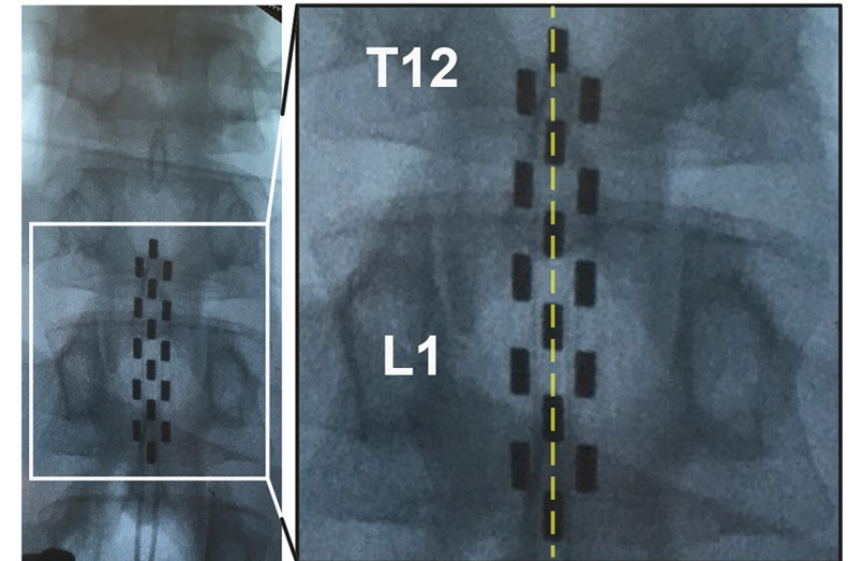
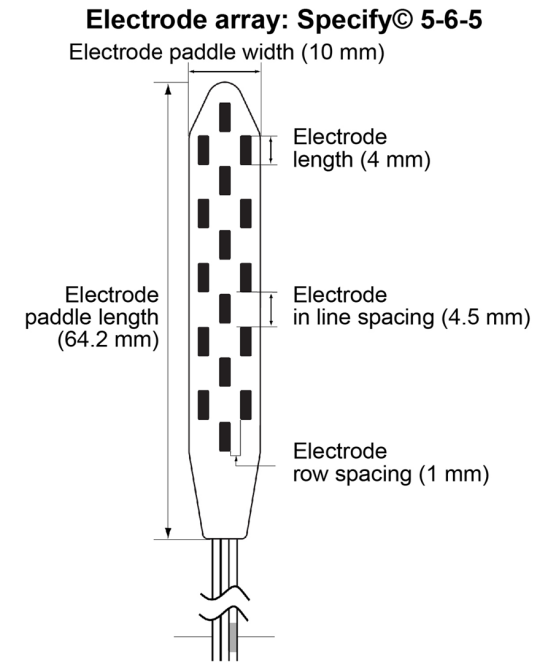
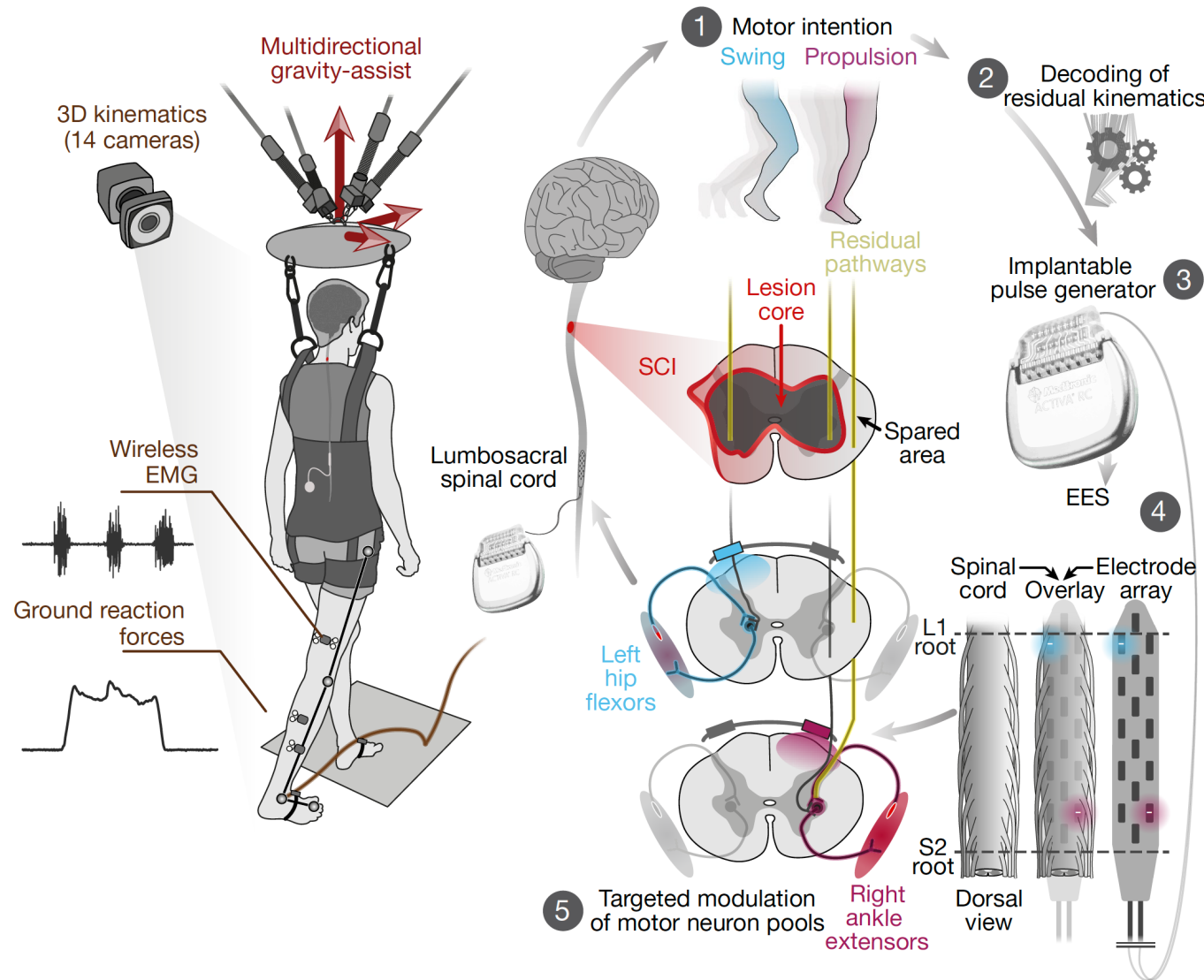
The electrical neuromodulation of the lumbar spinal cord is capable of reactivating the sensorimotor circuits.

EES:

- Recruits large sensory afferents, particularly proprioceptive feedback circuits.
- The recruitment of these pathways via electrical stimulation leads to the activation of the motor neurons through monosynaptic and polysynaptic proprioceptive circuits.
- Produces an excitatory drive that increases the general excitability of the spinal cord, which enables residual descending inputs to trigger volitional movements.



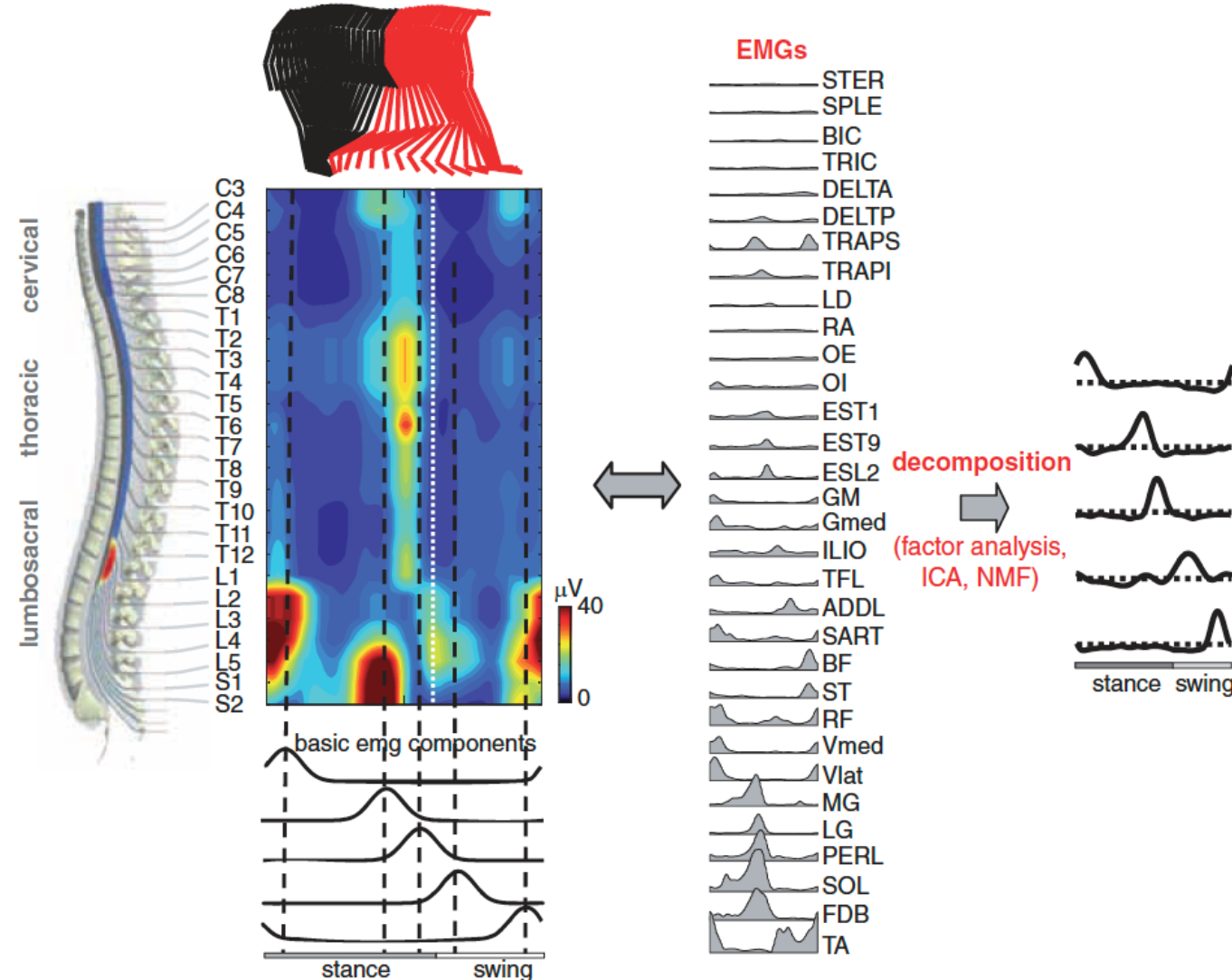
Epidural electrical stimulation (EES)



Epidural electrical stimulation (EES)

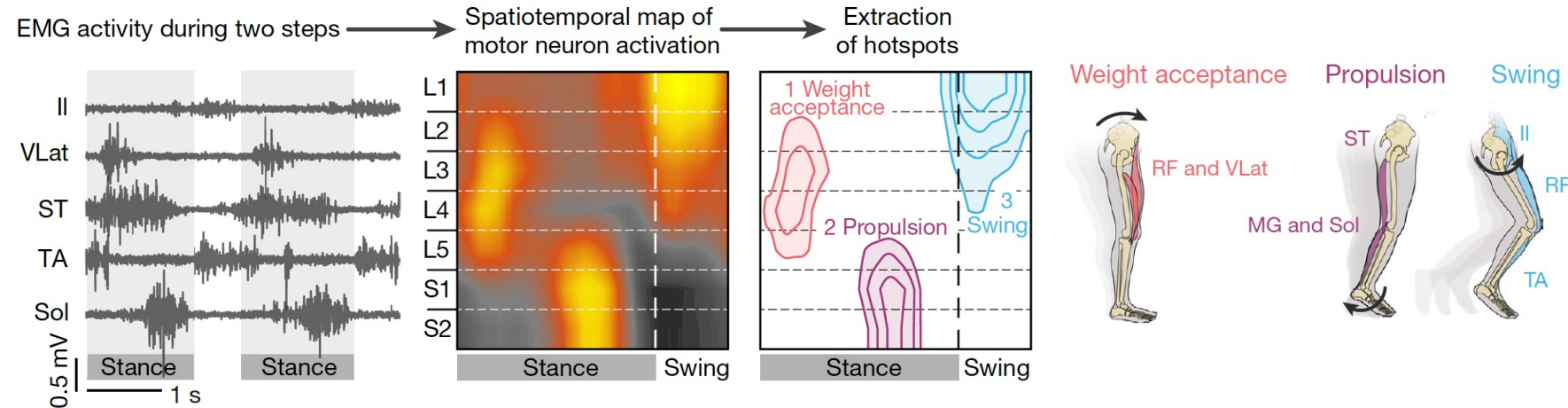
During **locomotion** there are MN activation hotspots that migrate during the progression of the gait cycle. Thus, continuous EES is intrinsically unfit to tune and control the modulation of specific MN pools during specific phases of the gait cycle.

- Since EES activates MNs through the modulation of proprioceptive feedback circuits that innervate distinct segments, spatially segregated motor pools may be accessed when delivering targeted stimulation of individual dorsal roots.
- The application of spatially selective EES bursts with a temporal sequence that reproduces the natural activation of motor neurons during locomotion allows a robust facilitation of extension and flexion muscle synergies.

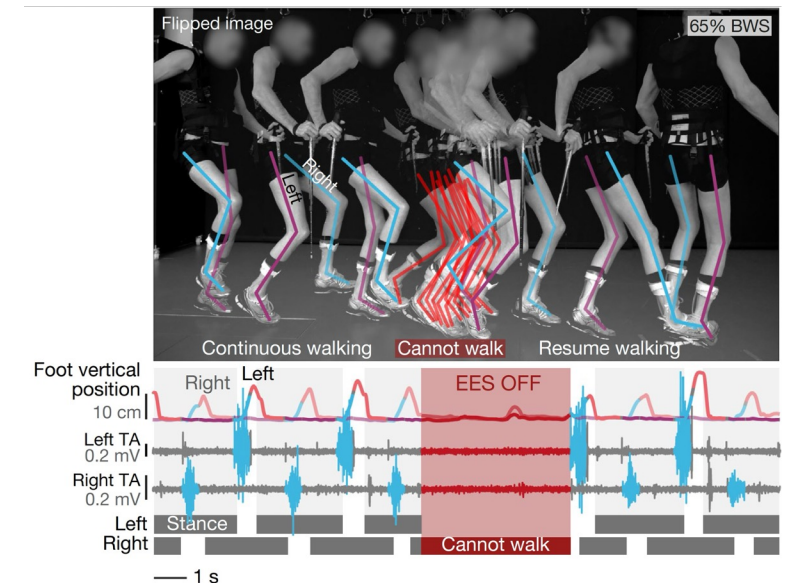
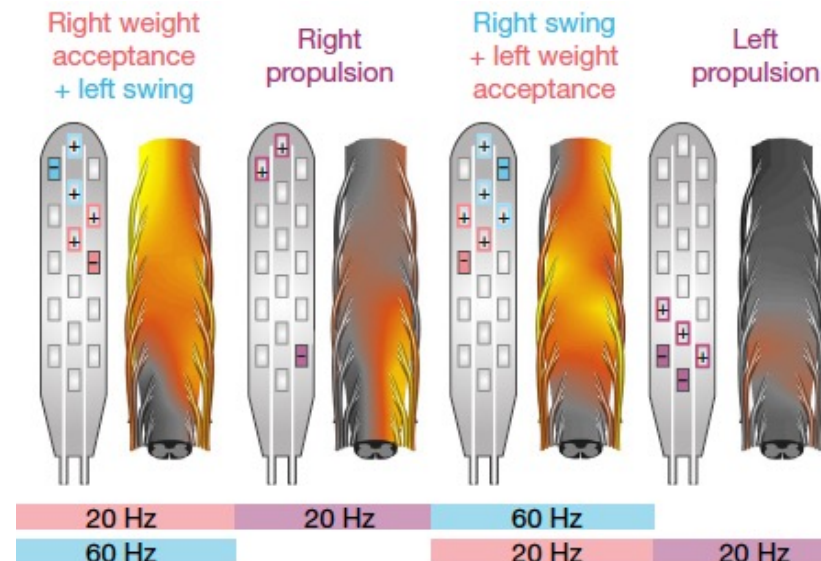


Epidural electrical stimulation (EES)

EMG activity during walking in healthy individuals shows a spatiotemporal map of motor neuron activation, revealing distinct activation hotspots. Each hotspot corresponds to a specific functional target.

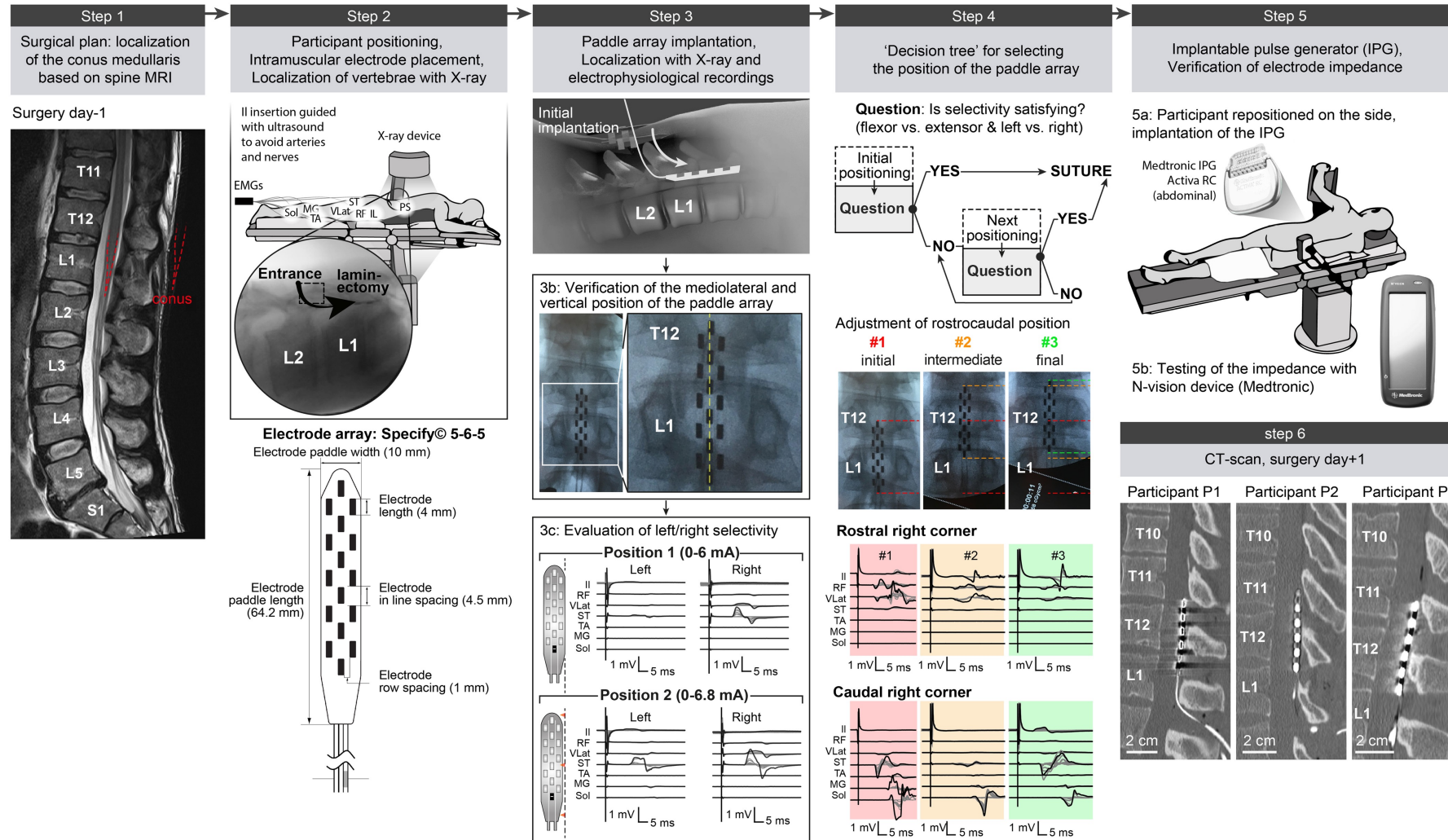


These maps are needed to create the spatiotemporal EES sequence to restore locomotion



Epidural electrical stimulation (EES)

Intra-operative characterization: to identify the optimal electrode placement

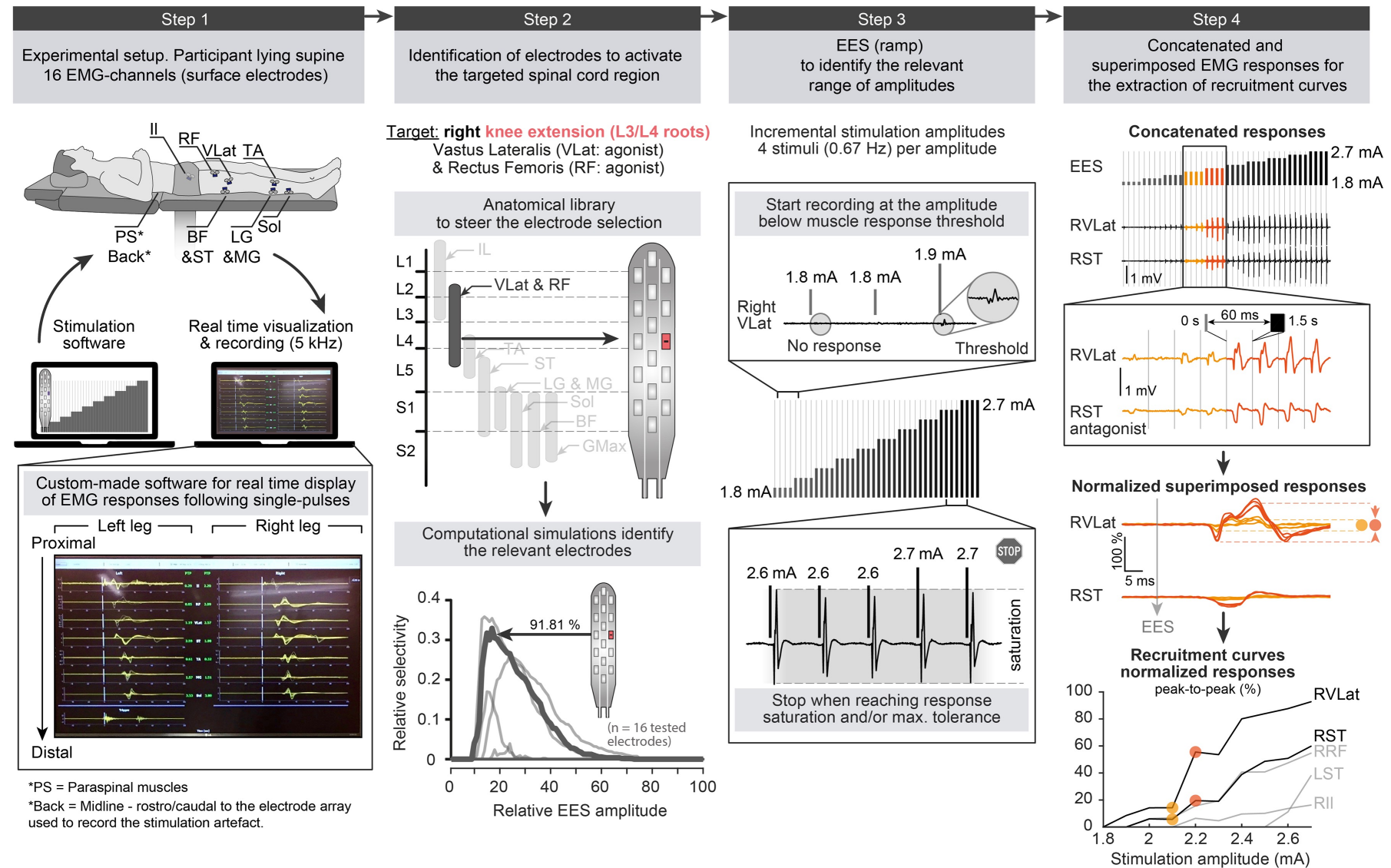


Epidural electrical stimulation (EES)

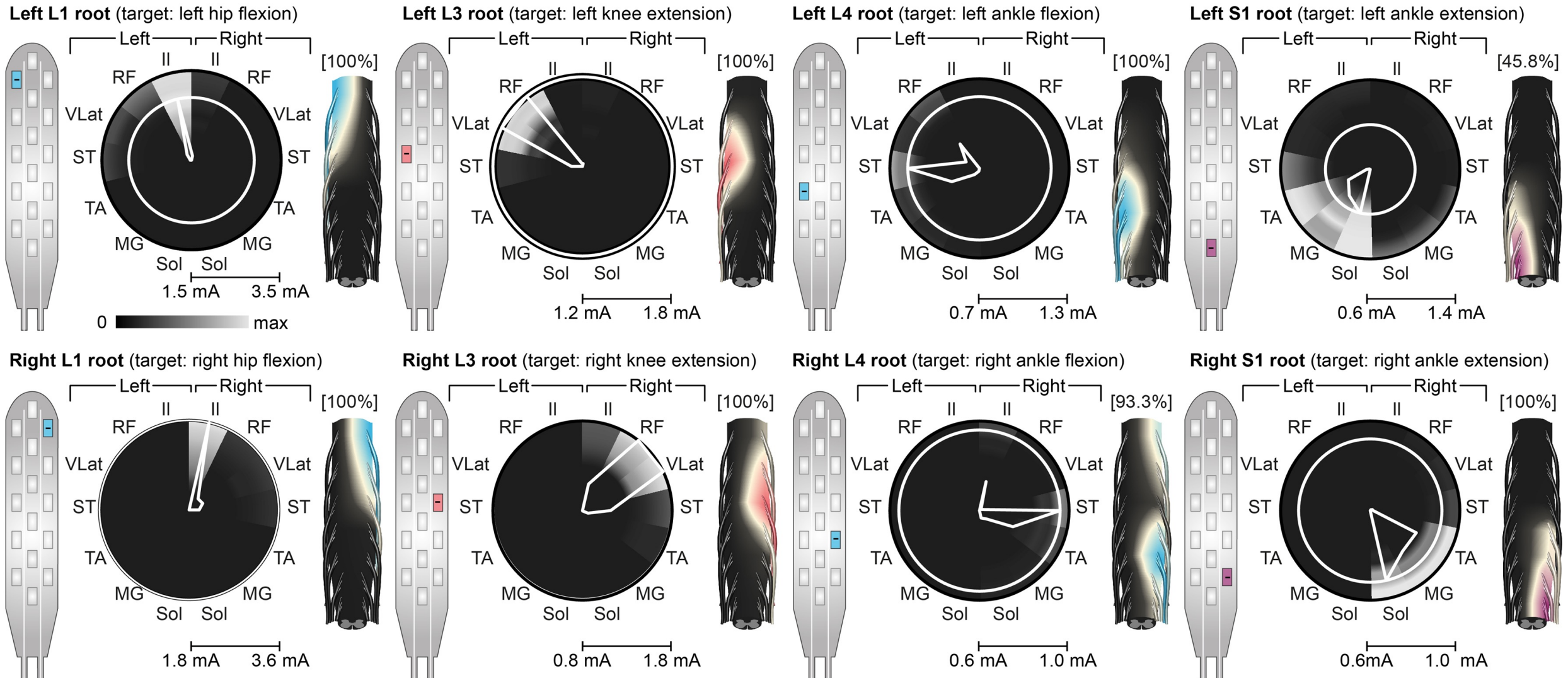
Post-operative characterization

Identifies the stimulation setup to restore walking and movements

1) Electrode mapping for recruitment curves.



Epidural electrical stimulation (EES)

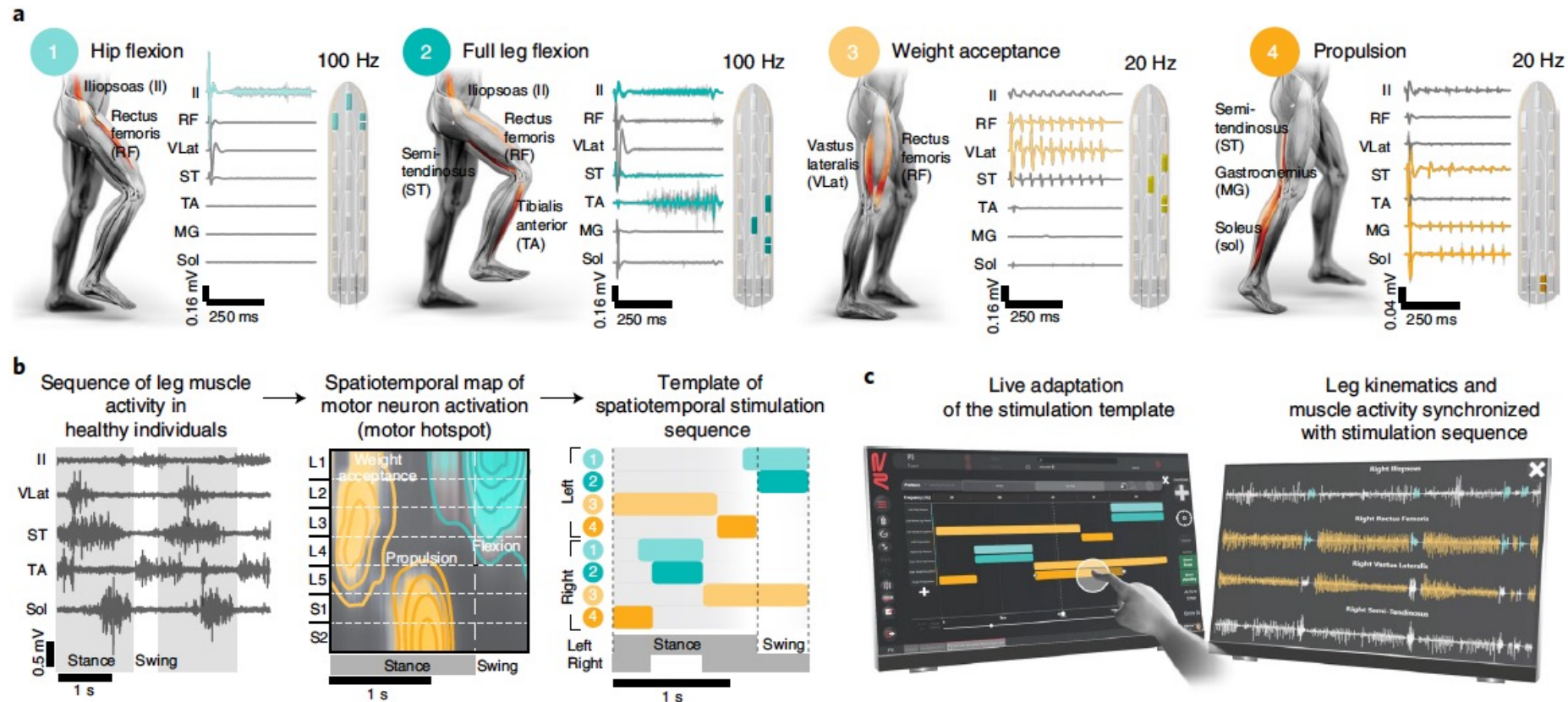


Epidural electrical stimulation (EES)

Post-operative characterization

Identifies the stimulation setup to restore walking and movements

- 1) Electrode mapping for recruitment curves.
- 2) Parameter tuning during movement and walking

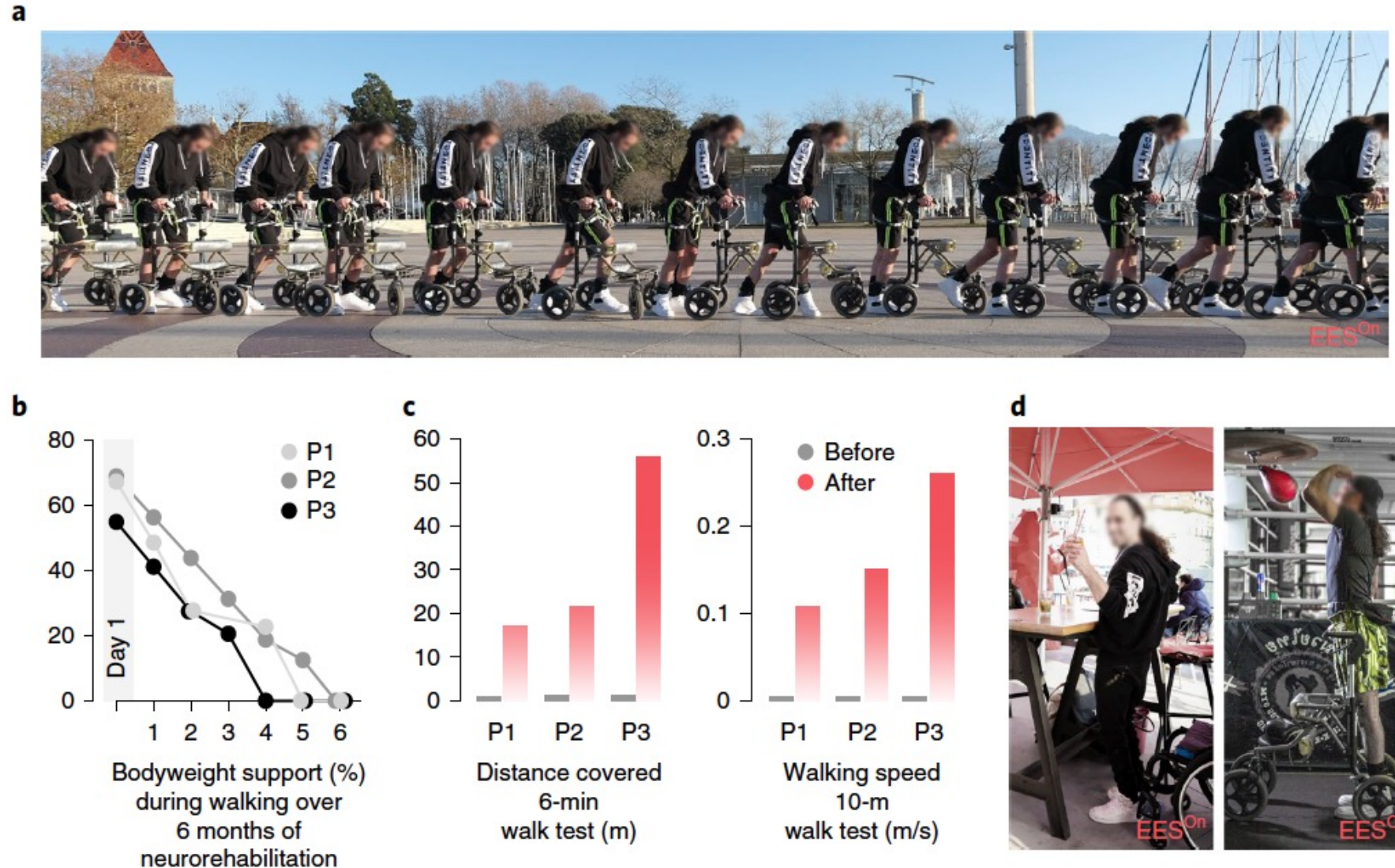


Epidural electrical stimulation (EES)

Post-operative characterization

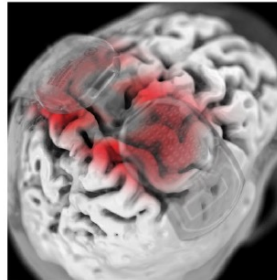
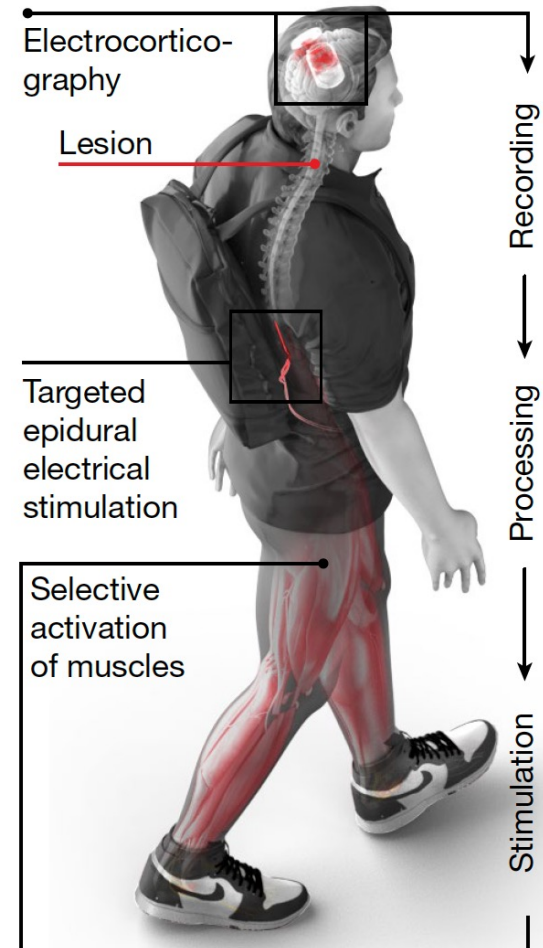
Identifies the stimulation setup to restore walking and movements

- 1) Electrode mapping for recruitment curves.
- 2) Parameter tuning during movement and walking
- 3) Follow-ups to keep track of improvements and fine-tune the parameters

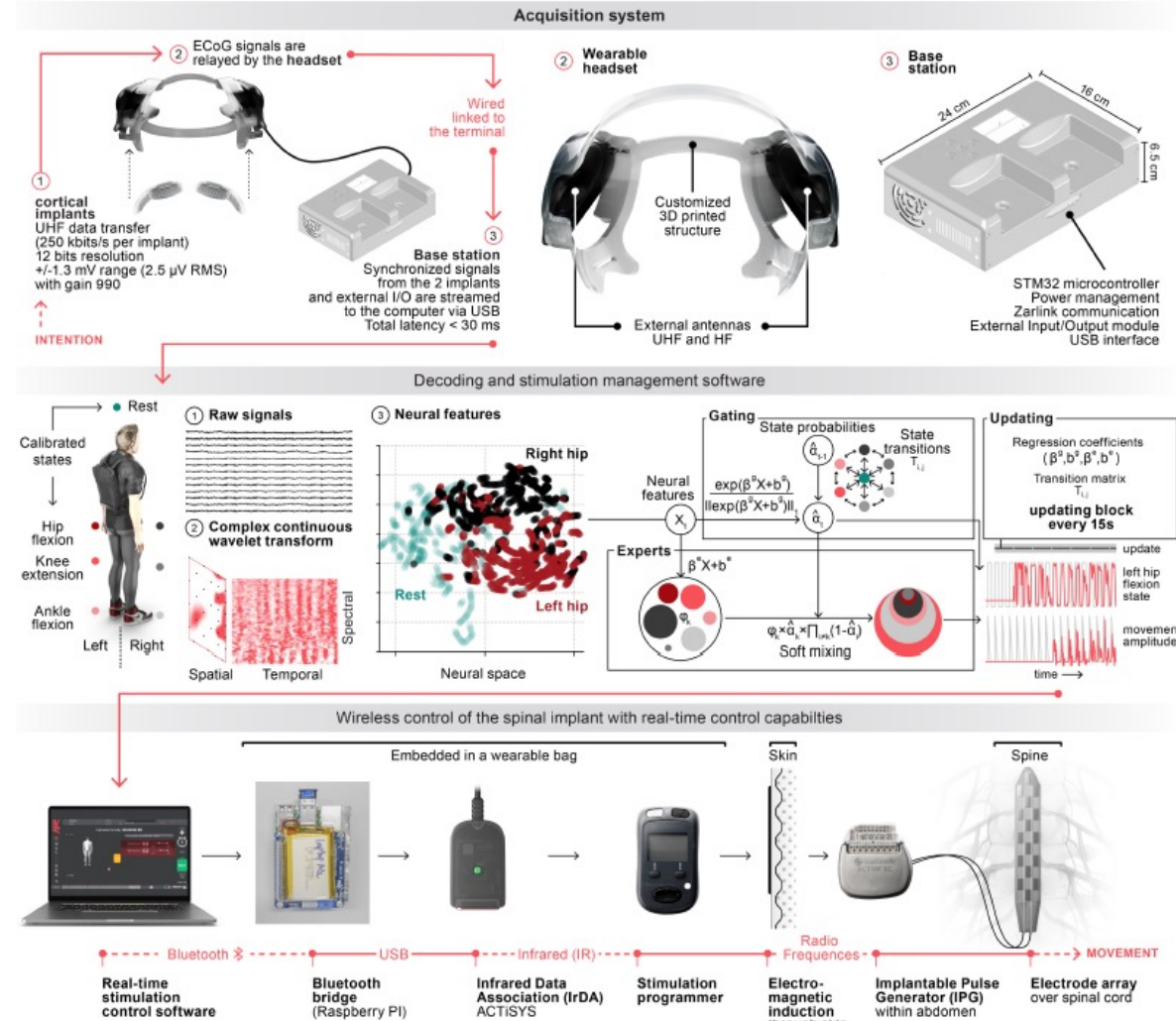
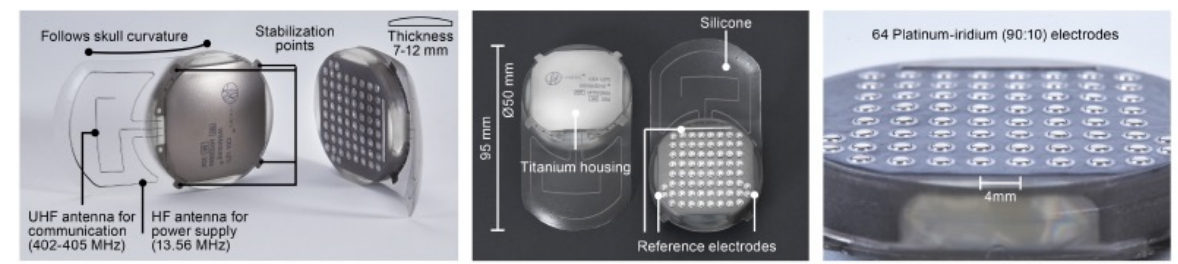
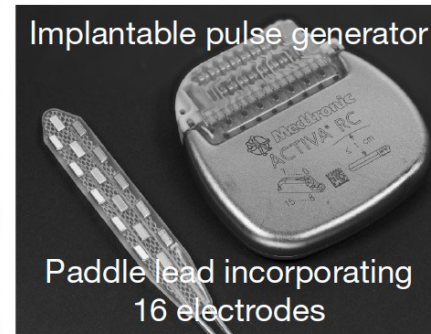
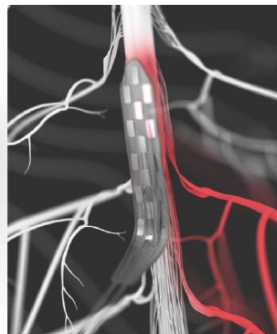


EES + Brain-spine interface

Cortical implants incorporating 2 × 64 channels

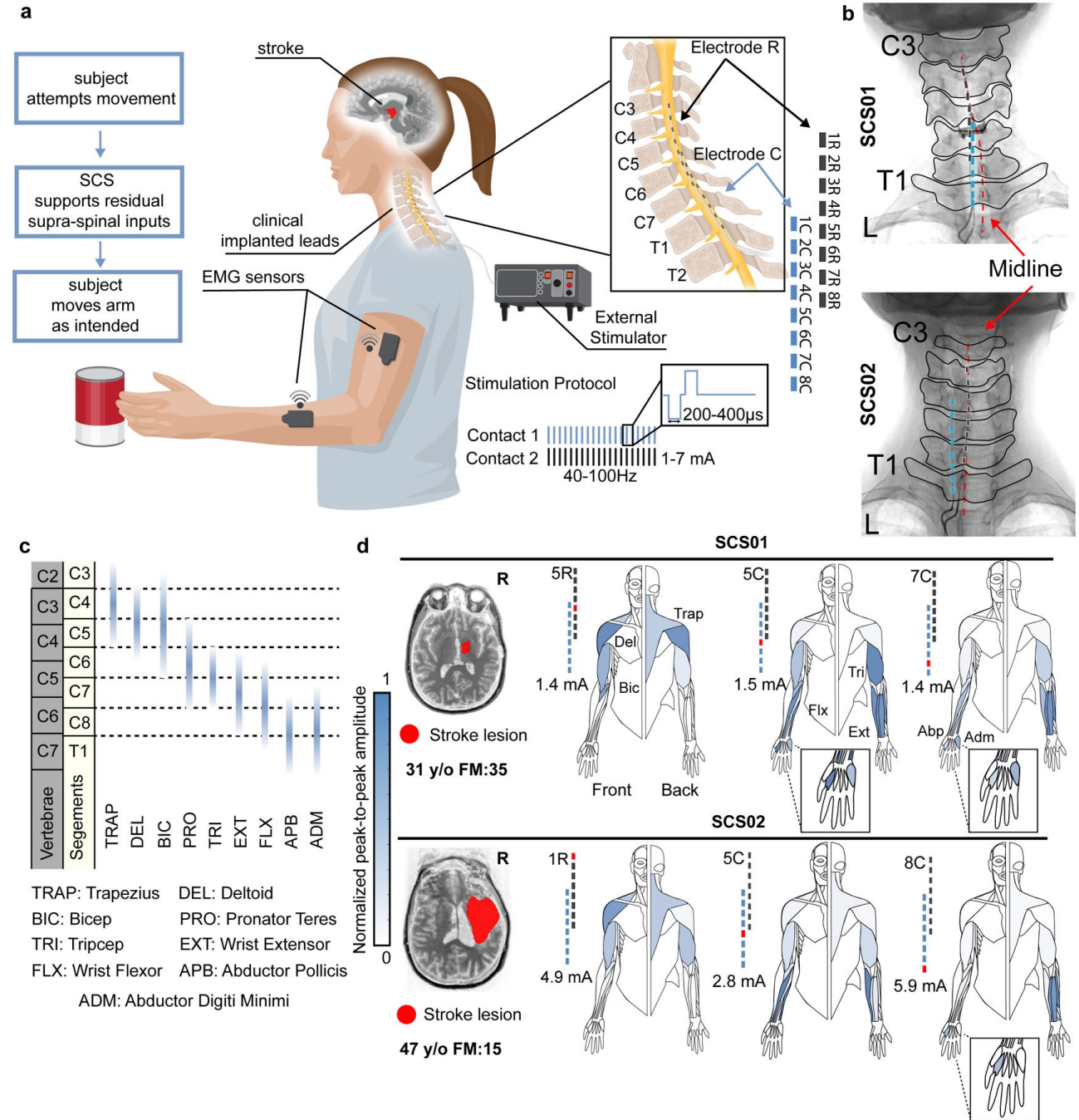


Wearable processing unit
 Receive neural data
 Extract spatial, temporal and spectral features to predict motor intentions
 Send updated stimulation commands

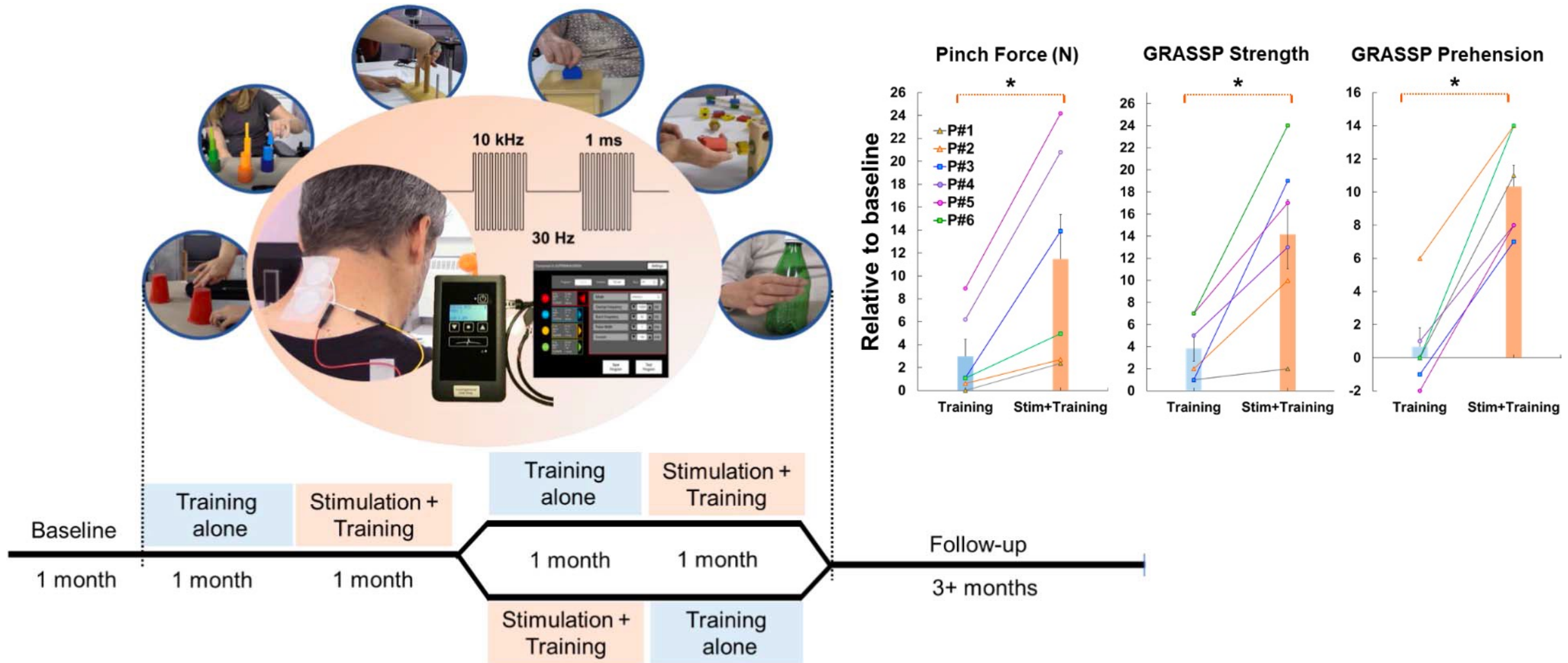


EES for upper limbs

- EES can improve the cortico-spinal control and restore arm and hand motor control in chronic post-stroke hemiparesis.
- Two participants implanted with linear leads targeting spinal roots C3 to T1 for 29 days.
- Grip force increased by +40% (SCS01) and +108% (SCS02). Speed increased by 30% to 40%.
- Enabled movements that were previously impossible without stimulation. Participants maintained some improvements even without stimulation.
- Preliminary data suggests SCS may be an effective assistive and restorative approach for upper limb recovery post-stroke, with no serious adverse events reported.



Transcutaneous spinal cord stimulation

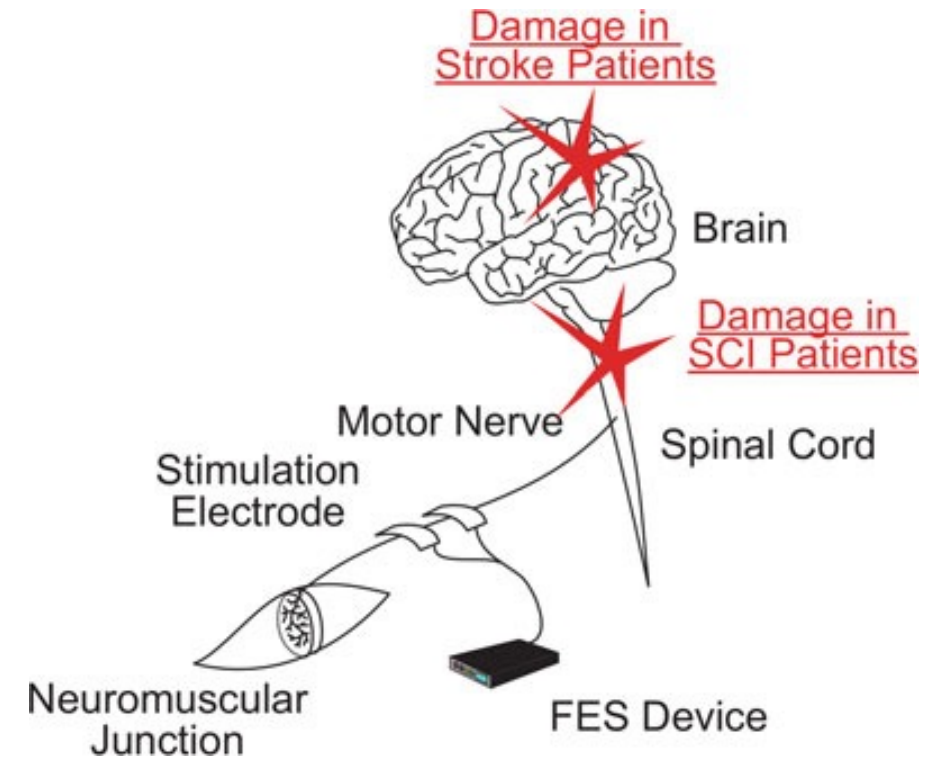


Functional electrical stimulation (FES)

- Individuals with stroke or SCI often struggle to voluntarily move various body parts and perform functions like sitting, standing, reaching, grasping, and bladder control. However, if the peripheral nerves, muscles, joints, and supporting tissues are intact, **electrical stimulation can be used to produce joint movements by contracting the muscles responsible for those actions.**
- This is called neuromuscular electrical stimulation (NMES). When NMES is applied in a **structured and patterned way to produce coordinated movements** like grasping, standing, or walking, rather than isolated muscle contractions, it is referred to as **functional electrical stimulation (FES).**
- Initially, FES was designed for neuroprostheses, intended to permanently substitute impaired functions like bladder control, grasping, and walking. In this setup, users activate the device whenever they want to perform a specific function.
- Recently, FES has been adapted as a therapy (**FES Therapy – FET**) to help restore voluntary motor functions like grasping, reaching, and walking. This approach is intended as a temporary intervention over several weeks or months, aiming to retrain the central nervous system to perform these functions independently, rather than relying on FES devices for life.

Functional electrical stimulation (FES)

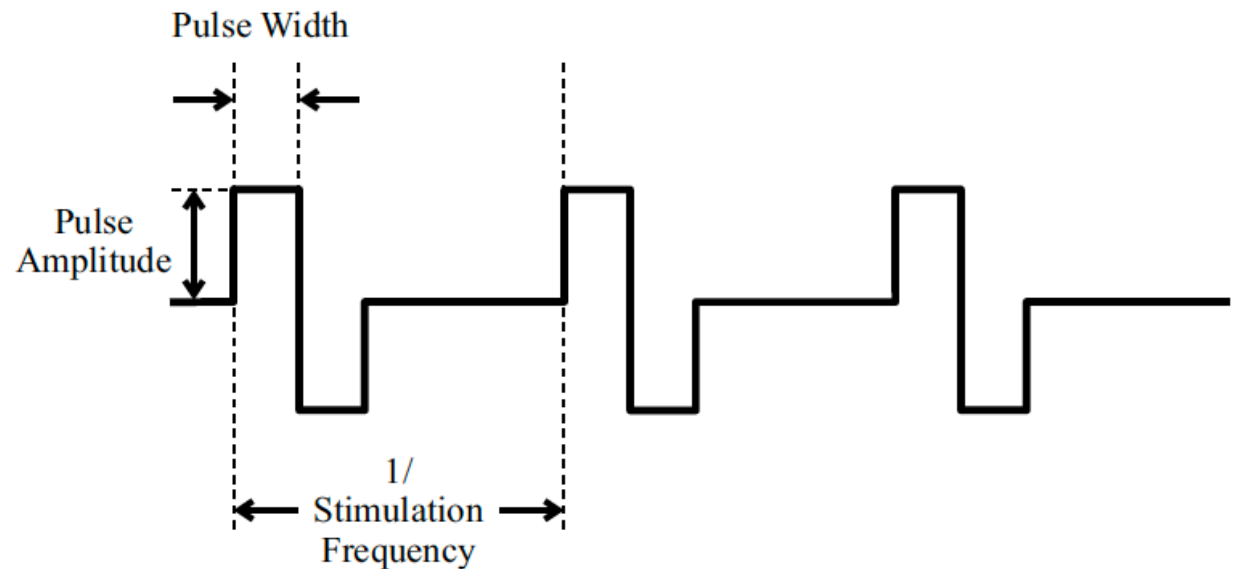
- In most cases, FES targets nerve activation. However, in some instances, FES can directly stimulate muscle fibers, particularly when peripheral nerves are severely damaged (such as in denervated muscles).
- The electrical stimulation generates action potentials in the motor neurons, which propagate along the motor neurons toward the muscle. When the action potentials reach the muscle, they cause the muscle to contract
- Most current FES systems stimulate the nerve trunk or nerve endings at the neuromuscular junction. This is largely because directly stimulating muscles requires significantly more energy – about three orders of magnitude more – making such systems harder to use in home and clinical settings.



Functional electrical stimulation (FES)

- In most FES applications, the electrical charge can stimulate both motor (efferent) nerves and sensory (afferent) nerves.
- Stimulating sensory nerves to trigger a reflex results in the coordinated contraction of one or more muscles. Activating the sensory system through reflex pathways is thought to play a significant role in enhancing motor function recovery after FES.
- Pulse width is usually set between 200-500 μs to cause effective muscle contractions. Frequency typically ranges from 20-50 Hz for sustained contractions; lower frequencies (<16 Hz) lead to unfused contractions, while higher frequencies (50-80 Hz) may cause faster muscle fatigue due to synchronized muscle fiber activation.

Functional Electrical Stimulation Parameters



Functional electrical stimulation (FES)

Surface electrodes: placed on the skin above the target nerve or muscle for non-invasive stimulation.

Pros:

- Non-invasive and easy to apply.
- Affordable and widely available.

Cons:

- Lower stimulation selectivity compared to subcutaneous options.
- Requires careful placement to maximize accuracy and reduce discomfort.
- Larger current needed, which can increase discomfort over time.

Percutaneous Electrodes: thin wires inserted through the skin into muscle tissue near the target nerve; typically for short-term use.

Pros:

- Higher selectivity compared to surface electrodes.
- Useful for temporary FES applications and can achieve effective muscle recruitment.
- Some centers have used them successfully for extended periods.

Cons:

- Prone to infection; requires strict care to minimize infection risks.
- Less practical for everyday, long-term use.
- Limited comfort for prolonged application due to external wires.

Implanted Electrodes: Permanently implanted under the skin near the target nerve, intended for long-term FES use.

Pros:

- High stimulation selectivity allows for precise muscle activation with lower stimulation amplitudes.
- Permanently embedded for continuous or regular use without reapplication.
- Reduced discomfort compared to surface and percutaneous electrodes.

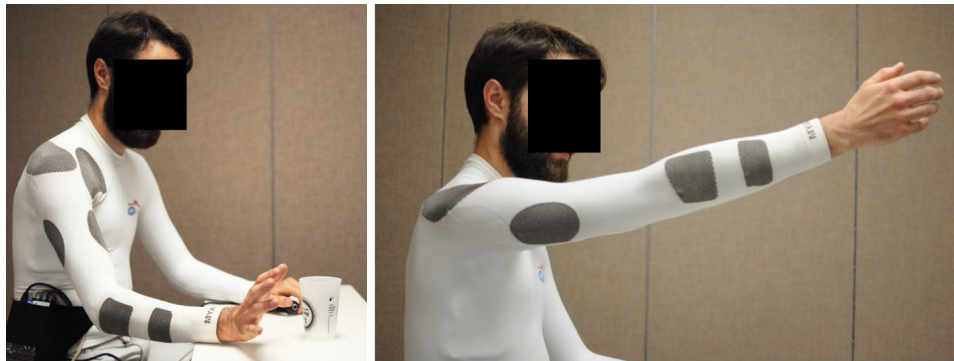
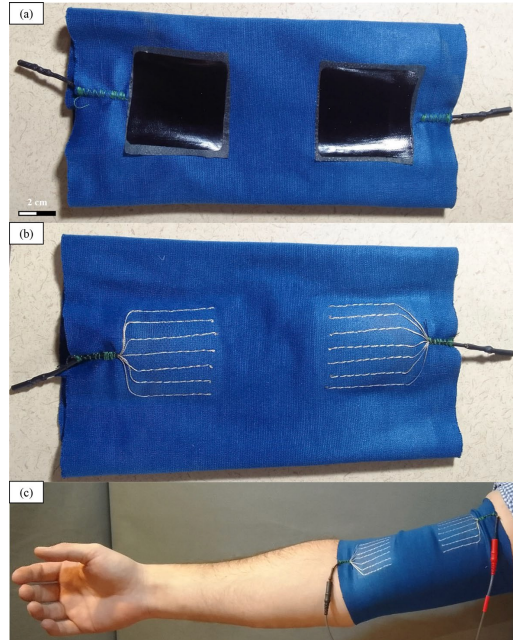
Cons:

- Invasive surgical procedure for implantation.
- Associated risks and complexities of long-term implant management.
- Higher cost and limited accessibility due to the need for surgical expertise.

Functional electrical stimulation (FES)

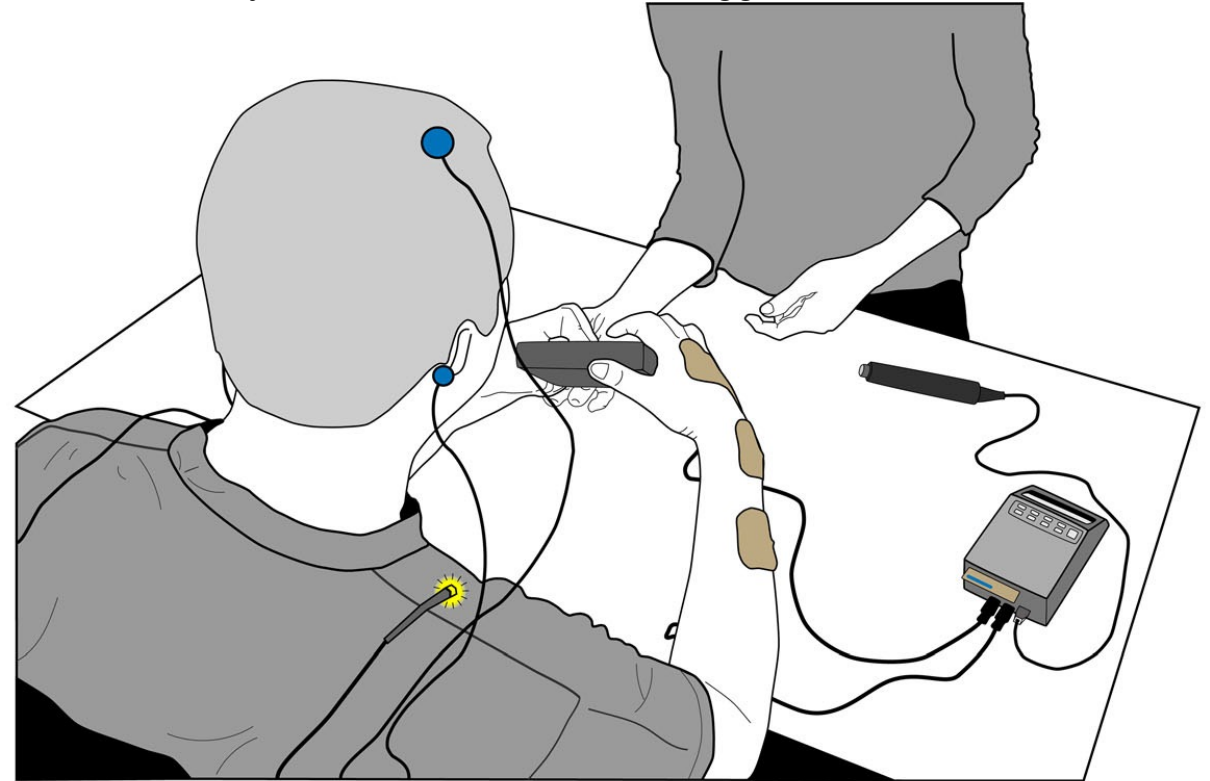
Smart FES Sleeve

The sleeve structure includes a carbon-based dry electrode integrated onto a textile substrate using thermal compression molding over a silver-plated conductive yarn matrix, enhancing both electrical connection and electrode fixation.



BCI-FET

Stimulation is initiated when the BCI detects an imagined or attempted movement. By using the BCI, there is greater assurance that the patient is voluntarily producing motor commands during therapy, as indicated by the EEG activity that determines when to trigger the stimulation.



Garnier, B., Marquez-Chin, M., DiNunzio, S., Iwasa, S. N., Saadatnia, Z., Naguib, H. E., & Popovic, M. R. (2024). A novel functional electrical stimulation sleeve based on textile-embedded dry electrodes. *BioMedical Engineering OnLine*, 23(1), 1-13.

Jovanovic, L. I., Popovic, M. R., & Marquez-Chin, C. (2021). KITE-BCI: A brain-computer interface system for functional electrical stimulation therapy. *The Journal of Spinal Cord Medicine*, 44(sup1), S203-S214.

2. Technologies for neurorehabilitation

2.1 Stimulation-based interventions

- Brain stimulations
- Spinal cord stimulations
- Functional electrical stimulation

2.2 Brain-computer interfaces (BCIs)

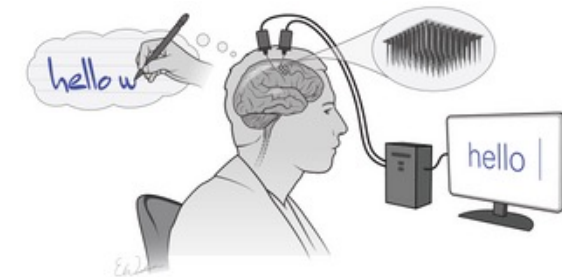
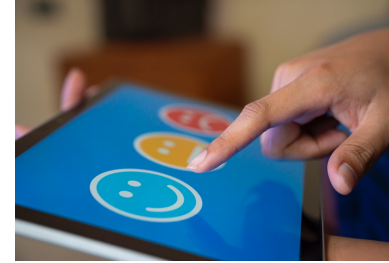
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- Framework and components of a neuroprosthesis
- Interfaces with the CNS and PNS

Human-machine interfaces (HMIs)

- A Human-Machine Interface (HMI) is the component of a device or software application that allows humans to operate and interact with machines.
- HMIs provide a bridge between the user and a machine's internal systems, enabling users to monitor, control, or command various functions.
- Examples of common HMIs:
 - Touchscreens
 - Keyboards and keypads
 - Graphical user interfaces (GUIs)
 - Voice-controlled interfaces
 - Gesture-based Interfaces
 - Augmented reality (AR) interfaces
 - Brain-computer interfaces (BCIs)

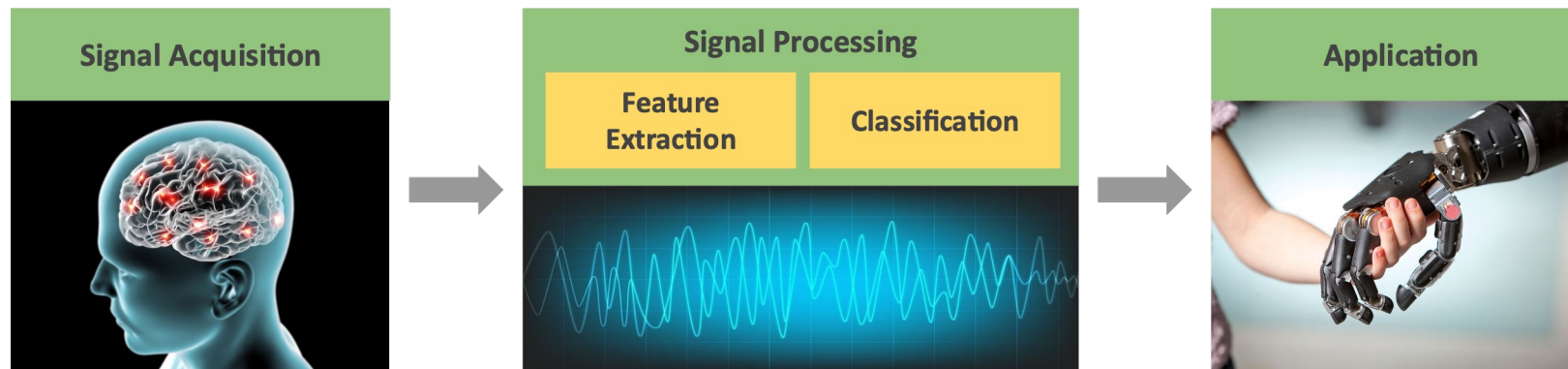
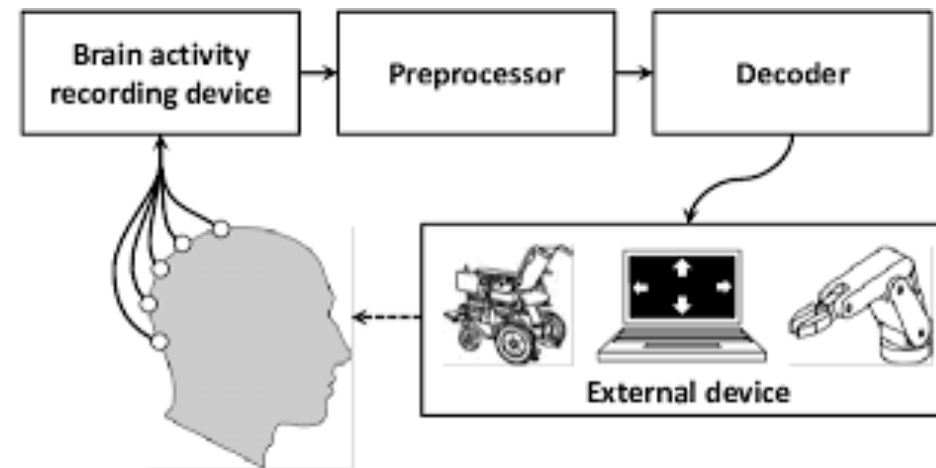


Brain-computer interfaces (BCIs)

- A BCI is a technology that enables **direct communication between the brain and an external device**, allowing for interaction without relying on the peripheral nervous system or muscle activity.
- BCIs **bypass the usual neuromuscular pathways**, instead using brain signals to control computers, robotic devices, or other assistive technologies. This approach is particularly useful for individuals with severe motor impairments who cannot use traditional input methods.
- Unlike other interfaces that depend on muscle signals or motor actions (i.e., Body-Machine Interfaces – BoMI), BCIs can function purely through brain activity, making them ideal for those with **spinal cord injuries, ALS, stroke, or other conditions affecting motor function**.
- BCIs are widely used in **assistive technology, neurorehabilitation**, gaming, and research, with applications ranging from controlling wheelchairs and prosthetics to enabling hands-free computer operation.

Brain-computer interfaces (BCIs)

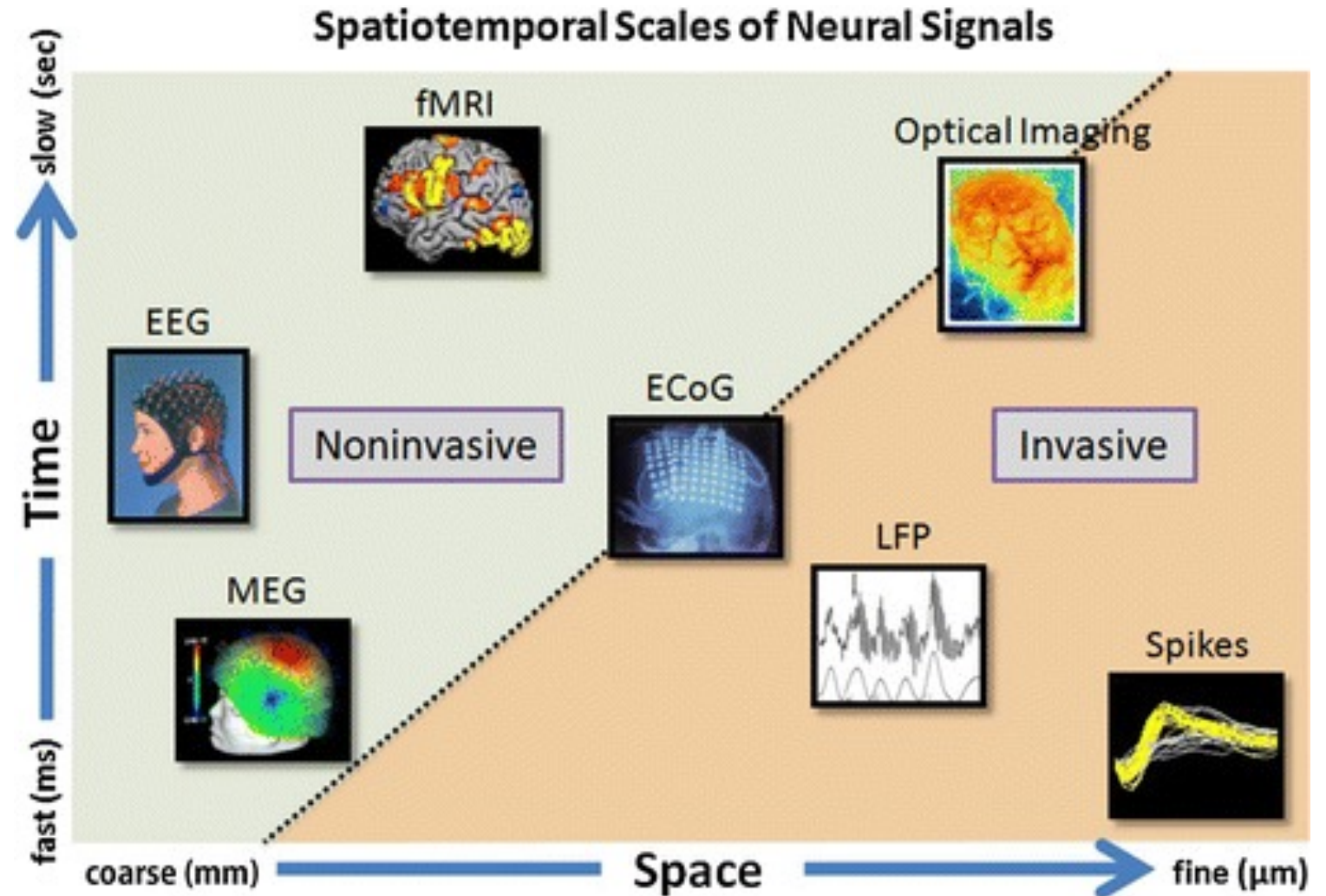
How BCIs work: Through sensors that detect and interpret electrical signals generated by the brain, BCIs decode the user's intentions and convert them into commands for external devices. This technology often uses electroencephalography (EEG) or implanted electrodes to capture and translate these brain signals.



Brain-computer interfaces (BCIs)

Control signals for BCIs

- Electroencephalography (EEG)
- Electrocorticography (ECoG)
- Local Field Potentials (LFPs)
- Functional Near-Infrared Spectroscopy (fNIRS)
- Magnetoencephalography (MEG)



Brain-computer interfaces (BCIs)

Invasive BCIs: require surgical implantation of electrodes directly onto or into brain tissue.

Electrocorticography (ECoG) - electrodes on the brain surface;
Intracortical microelectrodes - implanted within brain tissue.

Pros

- High spatial and temporal resolution
- Precise and direct brain signal access
- Greater control accuracy for complex tasks (e.g., robotic limbs, speech synthesis)

Cons

- Requires brain surgery, carrying significant medical risk
- Potential for infection, scarring, and device degradation over time
- Higher cost and limited to clinical/research settings

Non-Invasive BCIs: use external devices (e.g., electrodes on the scalp) to measure brain signals without surgery.

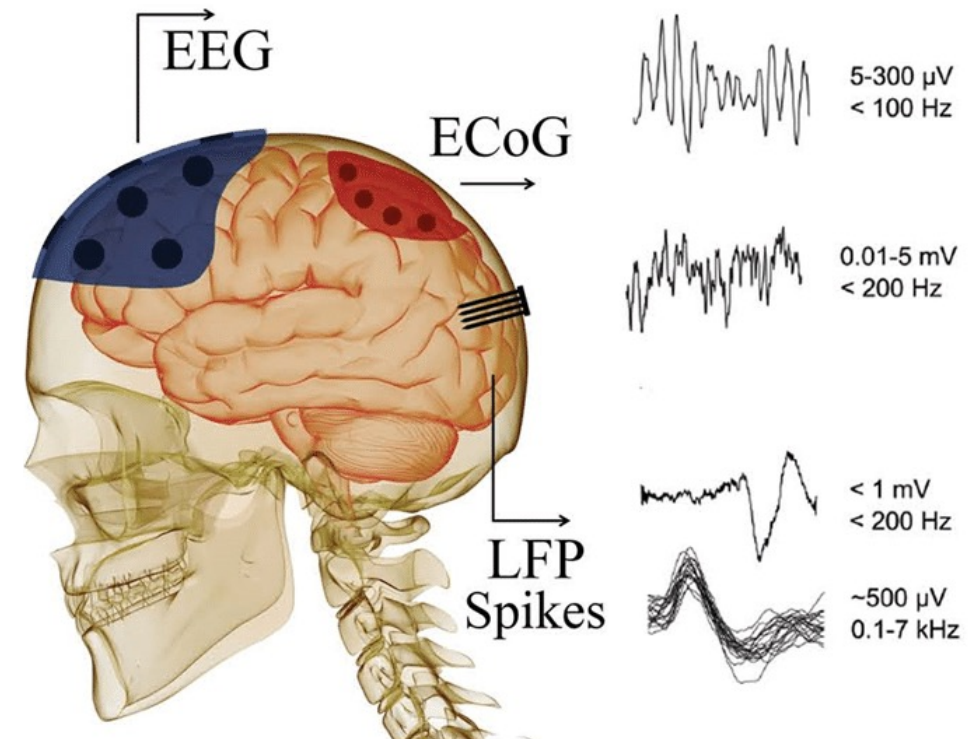
Electroencephalography (EEG) - records electrical activity through scalp electrodes;
Functional Near-Infrared Spectroscopy (fNIRS) - detects blood flow changes in the brain.

Pros

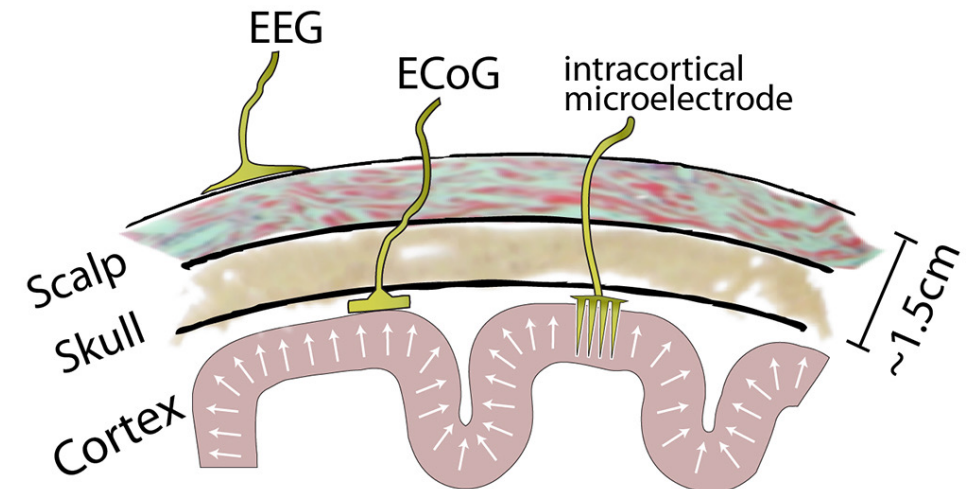
- Safe, painless, and easily repeatable without surgery
- Cost-effective and accessible for broader use
- Portable options suitable for home and rehabilitation settings

Cons

- Lower signal quality and spatial resolution due to interference from the skull and scalp
- Limited precision, making it challenging for complex tasks
- Generally slower response times compared to invasive BCIs

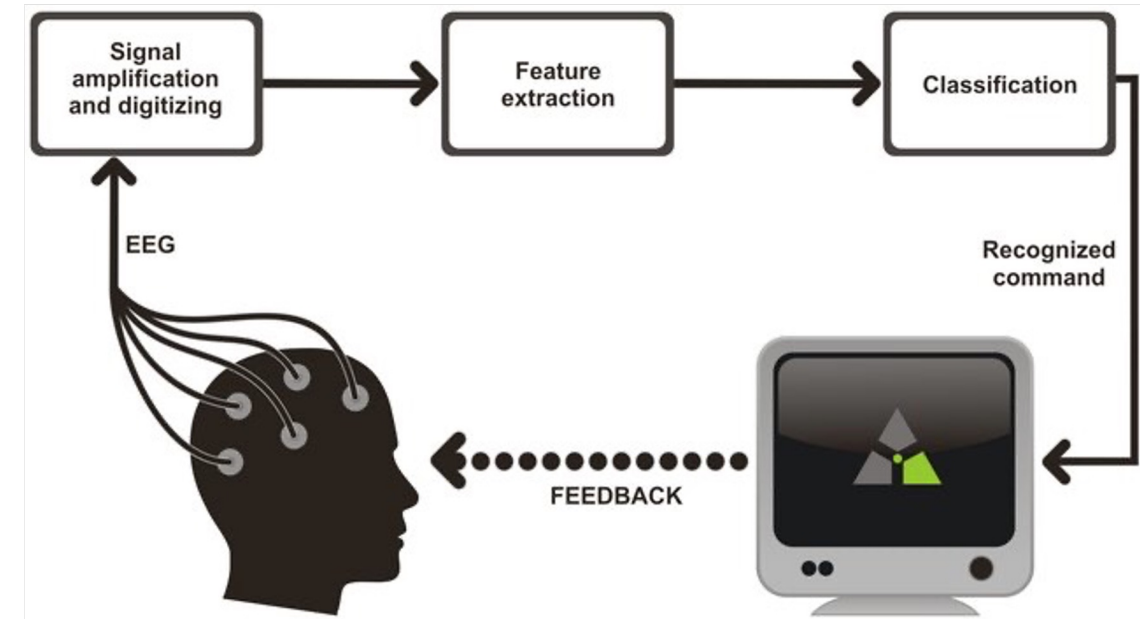
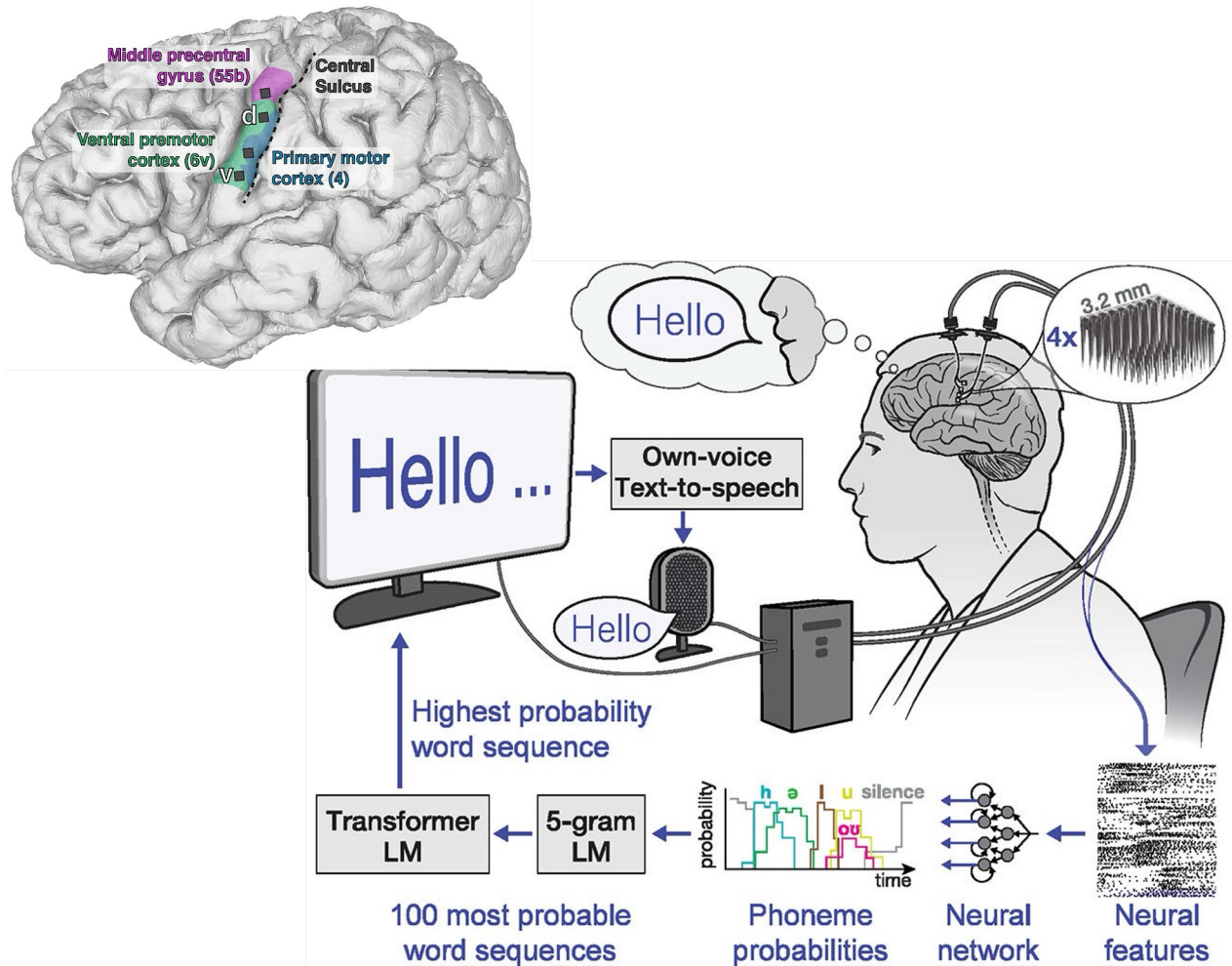


Lago & Cester 2017



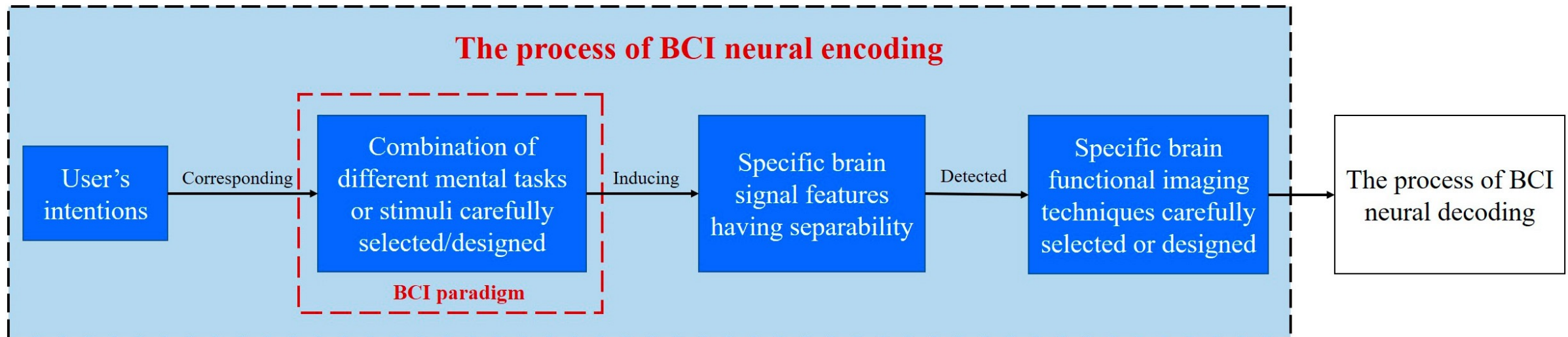
Miller et al. 2020

Brain-computer interfaces (BCIs)



BCI paradigms

A BCI paradigm consists of **specific mental tasks or external stimuli** chosen or designed by BCI developers **to represent a user's intentions**, captured by specific brain imaging techniques. The BCI paradigm's role is **to encode the user's intentions within brain signals**, creating a neural representation of those intentions. The brain imaging technology then detects this neural encoding, enabling the system to "read" or decode the user's intentions in subsequent steps.



BCI paradigms

Motor imagery paradigm

- It is based on imagining specific motor movements (e.g., imagining moving a hand or foot) without actually performing them. This activates certain brain areas, primarily in the motor cortex, that can be detected and interpreted.
- Often used in neurorehabilitation, robotic control, and assisting paralyzed users by enabling mental control over external devices.

P300 Event-related potential paradigm

- It uses the P300 wave, a distinct positive peak in EEG signals occurring around 300 milliseconds after a user sees or hears a relevant stimulus. Often used in "oddball" tasks where a target item is highlighted among non-targets.
- Commonly applied in communication systems, such as spelling devices, where users select letters or options on a screen by focusing on them.

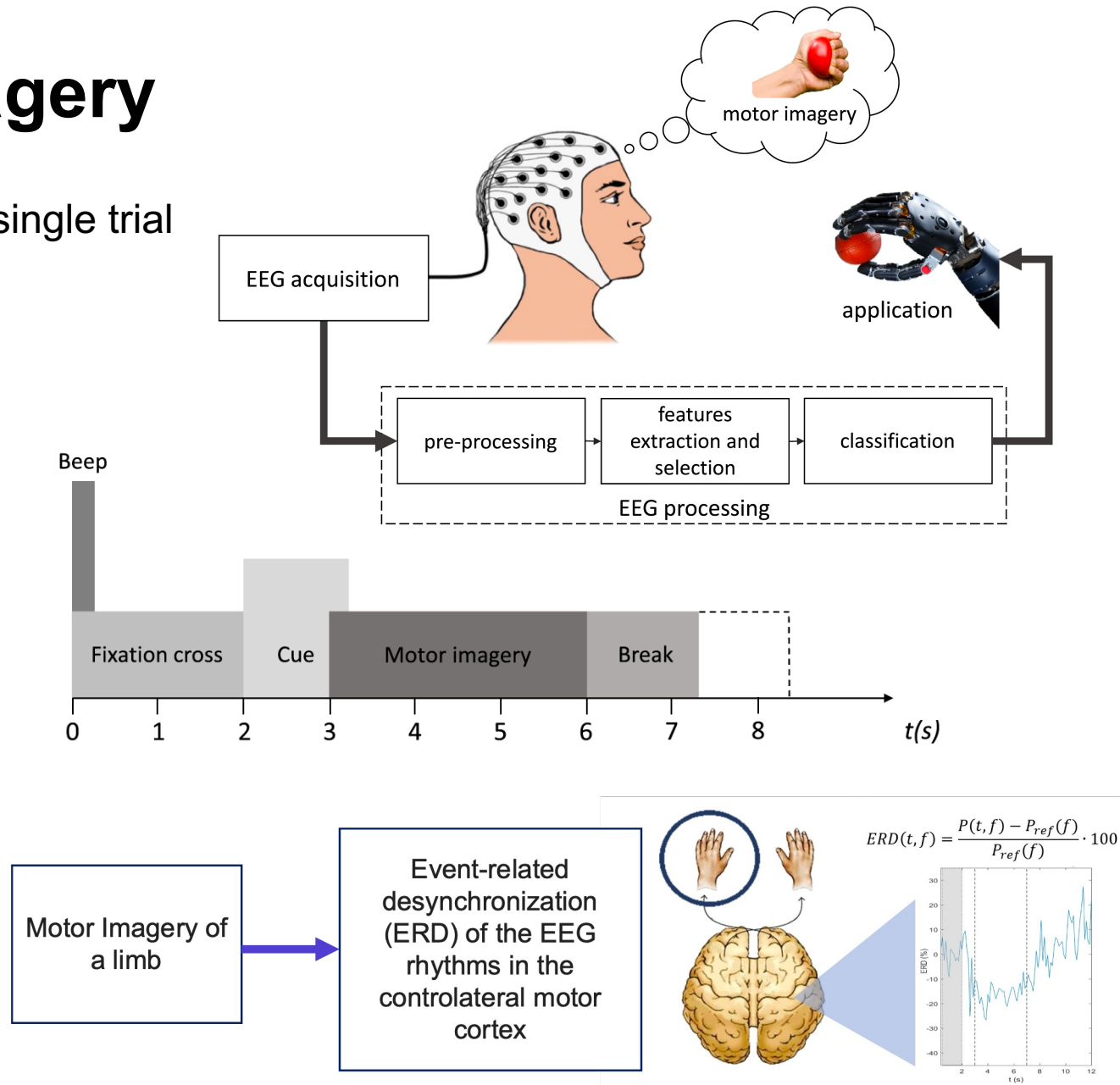
Steady-state visual evoked potential (SSVEP) paradigm

- It involves presenting the user with visual stimuli flashing at specific frequencies. When a user focuses on a particular flashing stimulus, the brain synchronizes at that frequency, which the system detects.
- Frequently used for device control and visual selection, such as controlling a cursor or selecting items on a computer screen.

BCI paradigms – Motor imagery

In a synchronous motor imagery (MI) experiment a single trial consists of four main stages:

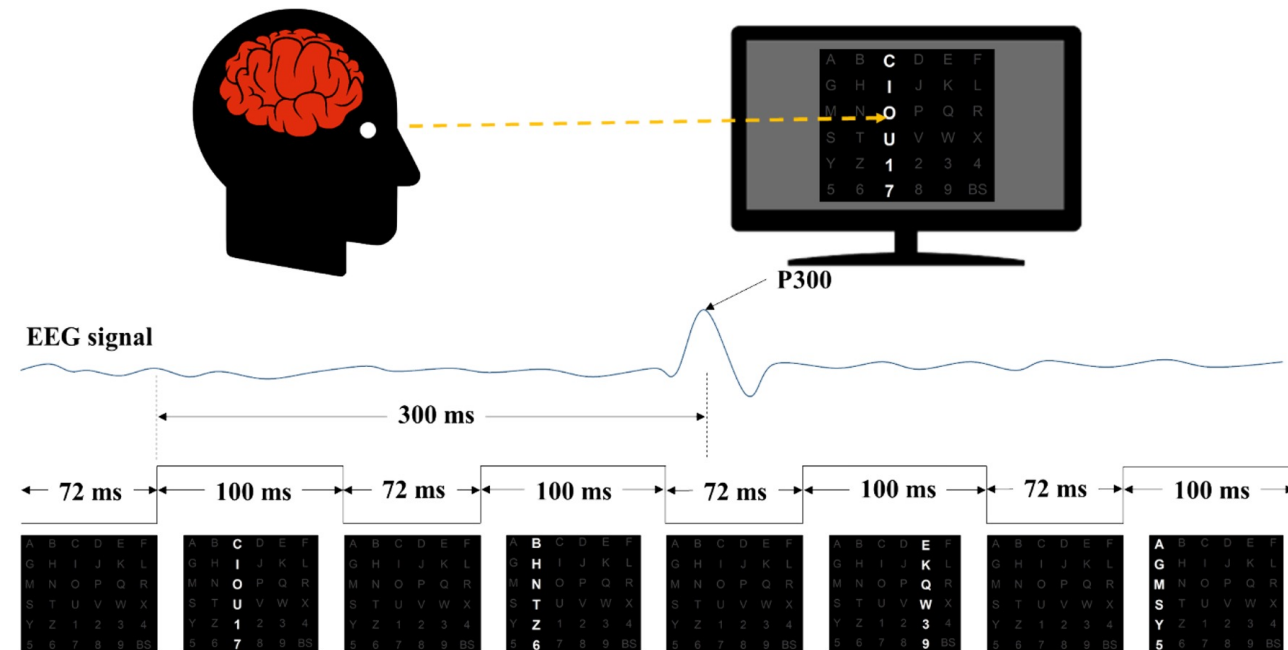
- **Relax phase:** the user begins by relaxing, often triggered by an acoustic signal, while **focusing on a fixation cross** on the screen to minimize eye movement.
- **Cue phase:** A prompt appears, indicating the specific imagination task to perform shortly.
- **MI task phase:** The user performs the motor imagery task, typically lasting 3–5 seconds.
- **Ending relax phase:** A short, random-duration break follows to prevent the user from adapting to specific timing patterns.



BCI paradigms – P300

A **P300 speller BCI** is a BCI system that allows users to spell out words and communicate by detecting brain responses, specifically the P300 wave in EEG signals.

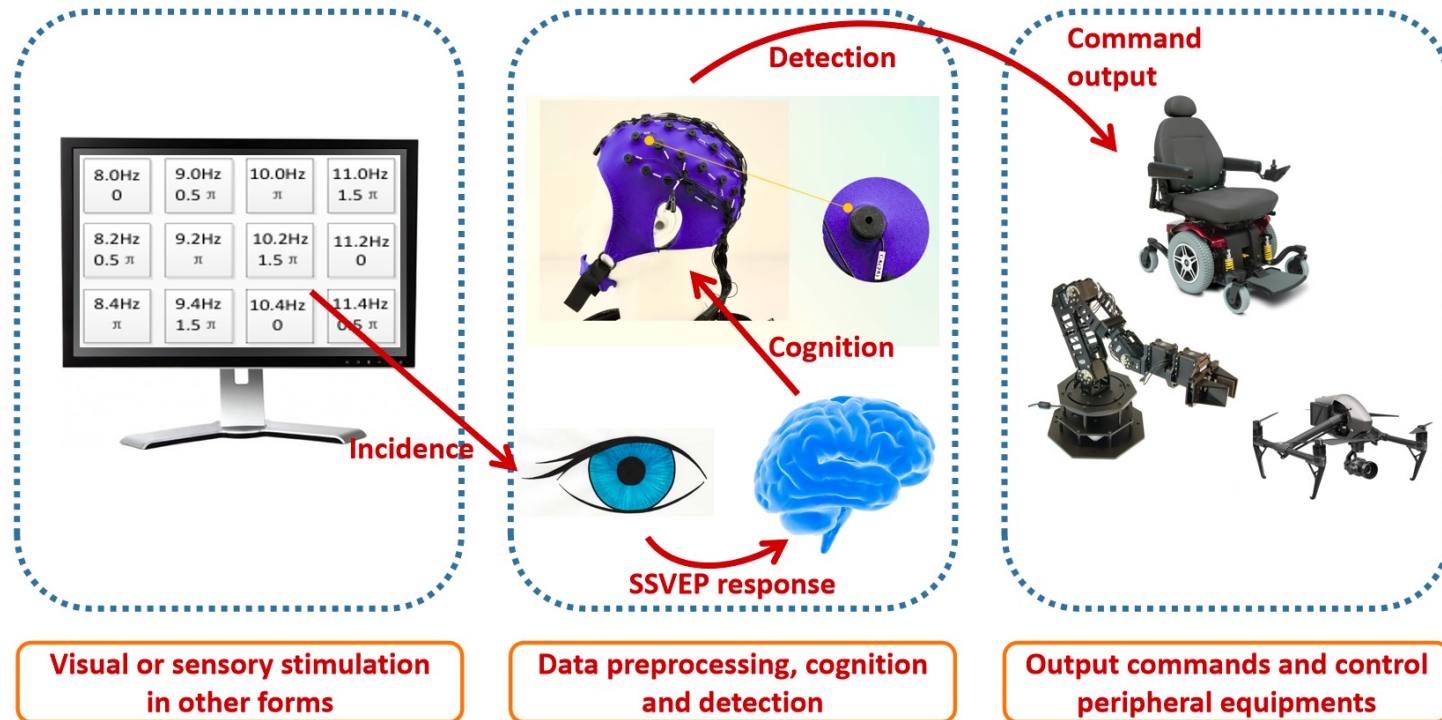
- An interface displays a grid of letters on a screen. The user focuses the attention on the letter to select.
- Rows and columns of the grid flash in random sequences. When the row or column containing the target letter flashes, it elicits a unique brain response in the user.
- The brain generates a specific signal, known as the P300 wave, ~300 ms after seeing the target flash. This wave is an ERP that occurs only when the user recognizes a stimulus of interest (in this case, the target letter).
- The BCI system measures EEG signals and uses algorithms to detect the P300 wave. By identifying which row and column produced the P300 response, the system pinpoints the target letter.
- The detected letter is then displayed as the next character in the message or word that the user is spelling.



BCI paradigms – SSVEP

SSVEP refers to a brain wave response that occurs when a person focuses on a visual stimulus that is flickering at a constant frequency.

- The brain synchronizes its electrical activity to the frequency of the stimulus, generating an EEG signal that can be detected and decoded by a BCI system.
- A person is presented with multiple visual stimuli (e.g., flashing squares, lights, or targets) at different frequencies.
- The user focuses on one target, and the brain responds to the specific frequency of that stimulus.
- The BCI system detects the frequency of the SSVEP signal to identify the user's choice, such as selecting a letter or controlling a device.



Brain-computer interfaces (BCIs)

Non-invasive EEG devices

g.Tec



(Unicorn)

Neuroelectrics



(Enobio)

Advanced Brain Monitoring



(B-Alert)

Emotiv



(EPOC)

InteraXon



(Muse Headband)

NeuroSky



(MindWave)

OpenBCI



(Ultracortex "Mark IV")

Brain Product



(LiveAmp)

Compumedic Neuroscan



(Quik-Cap – SynAmps)

EEGSmart



(Touch)

Nihon Kohden



(Vital EEG)

Cognixion

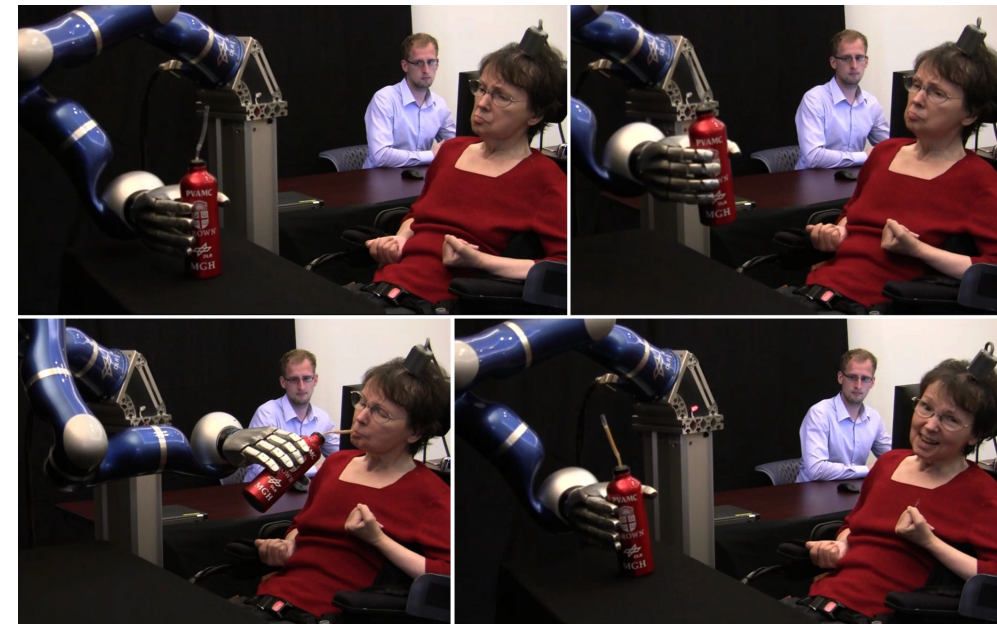
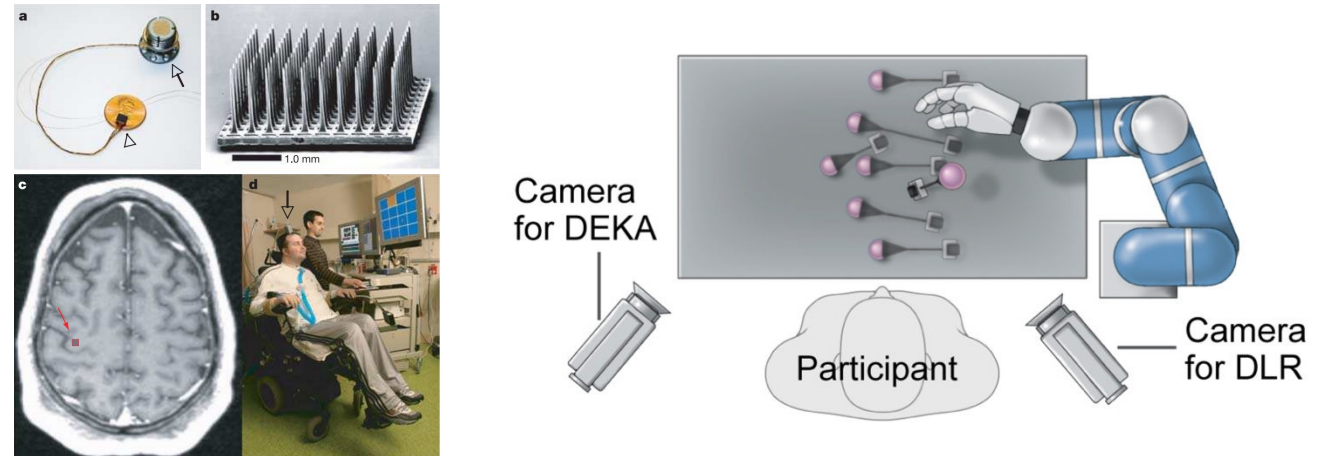


(Cognixion ONE)

Brain-computer interfaces (BCIs)

BCIs for control

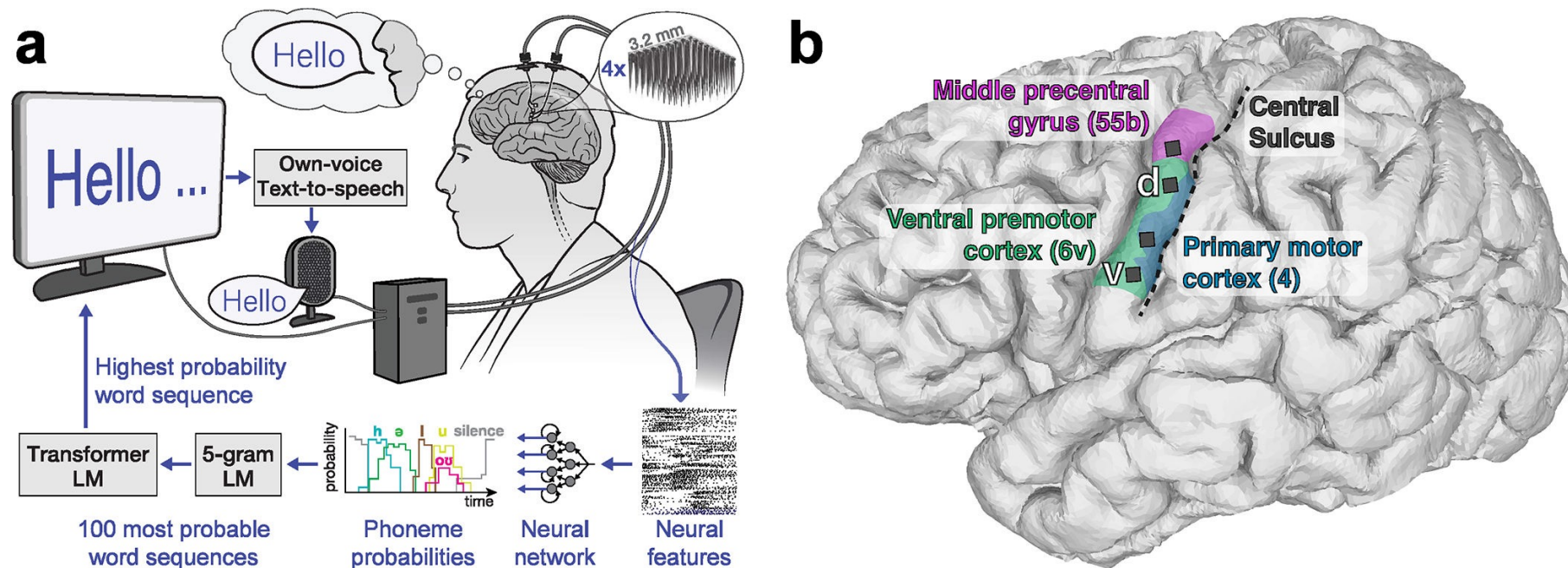
- This BCI enables participants to control an arm using signals decoded from a small group of motor cortex neurons recorded by a 96-channel microelectrode array.
- Even without explicit training, participants could control a robotic arm to perform tasks such as drinking coffee.
- Although the robotic arm's movements were slower and less precise than an able-bodied person's, the results show that people with tetraplegia, even years after spinal cord injury, can regain control over complex devices through direct neural signal decoding.



Brain-computer interfaces (BCIs)

BCIs for speech and communication

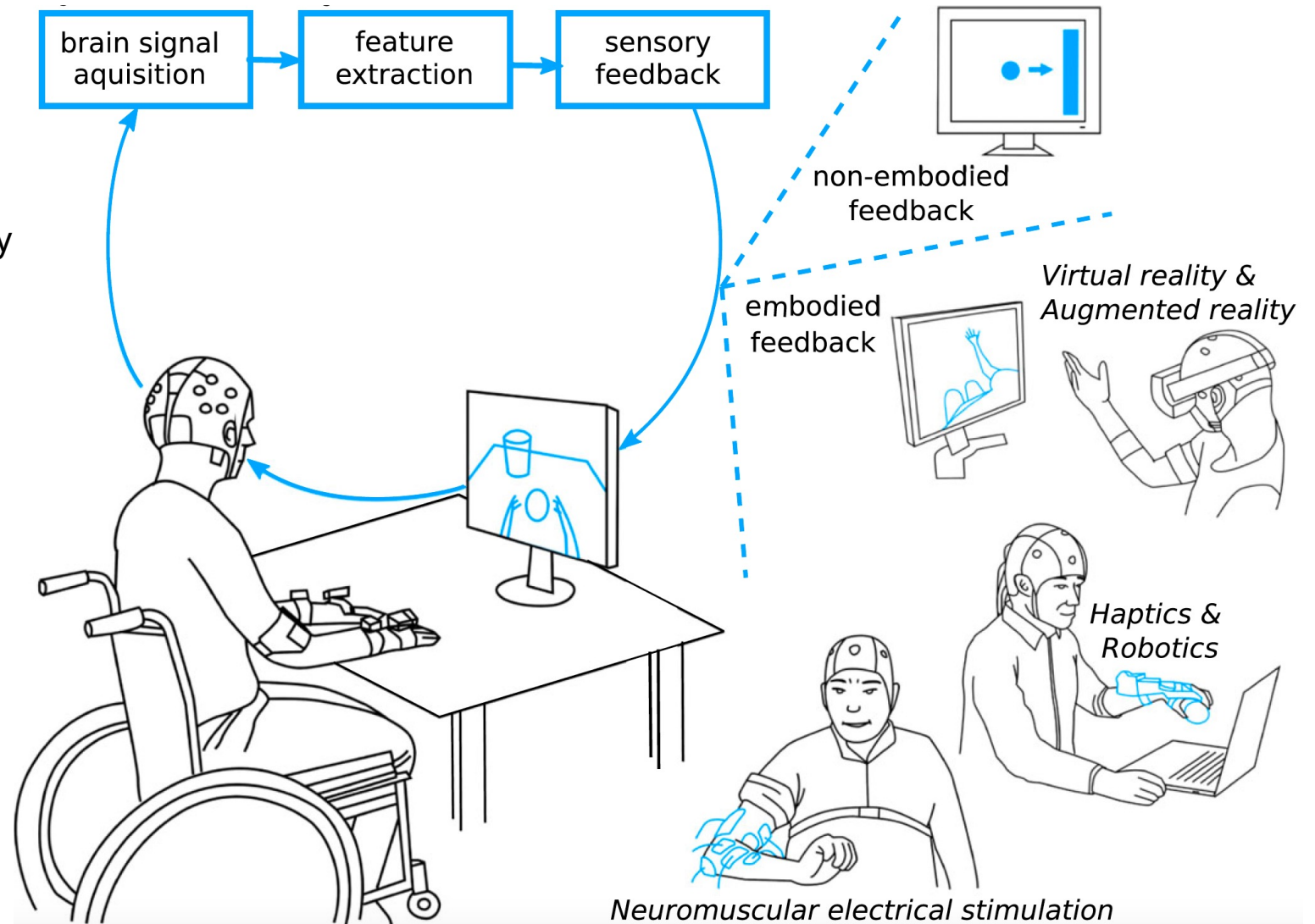
The brain-to-text speech neuroprosthesis system measures cortical neural activity using four 64-electrode arrays. Machine learning algorithms process this neural data to decode it into an English phoneme every 80 ms. The microelectrode arrays are strategically placed on a 3D reconstruction of the participant's brain, with their locations (shown as gray squares) aligned to specific cortical regions. These regions are mapped using MRI scans from the Human Connectome Project, ensuring precise alignment before implantation.



Brain-computer interfaces (BCIs)

BCIs for neurorehabilitation

- BCIs in stroke rehabilitation promote neuroplasticity and motor recovery.
- BCIs regulate neurophysiological activity to facilitate cortical reorganization.
- They can stimulate activity in either the ipsilesional or contralesional hemisphere, depending on the patient's condition.
- Recovery mechanisms include Hebbian learning, error-based learning, and reward-based learning.
- BCI training increases motor cortex excitability, corticospinal tract activity, and spinal motoneuron pool excitability.
- Real-time feedback (e.g., robotic orthoses or virtual avatars) helps patients control sensorimotor oscillations, enhancing recovery.



2. Technologies for neurorehabilitation

2.1 Stimulation-based interventions

- Brain stimulations
- Spinal cord stimulations
- Functional electrical stimulation

2.2 Brain-computer interfaces (BCIs)

- Overview of BCIs
- BCI paradigms
- Types of BCIs and applications

2.3 Neural interface technologies

- Framework and components of a neuroprosthesis
- Interfaces with the CNS and PNS

Neuroprostheses

- Neurological disorders cause lasting sensory-motor deficits, impacting quality of life and health care systems.
- The effects of these disorders are projected to increase with an aging population.
- In Europe, roughly 1.1 million people experience a stroke annually, often resulting in severe motor deficits, especially in upper limb function.
- Other conditions like Parkinson's disease, multiple sclerosis, and injuries (e.g., spinal cord or brain injury) also contribute to motor impairments, reducing patient independence.
- Neuroprosthetics offer promising solutions to improve quality of life for individuals with these limitations.

A neuroprosthesis is a technology designed to **interact with the nervous system** through implantable or noninvasive interfaces, aimed at restoring lost neurological functions. Neuroprostheses can bridge different levels of the nervous system that have been disrupted by injury or neurological disorders.

Neuroprostheses

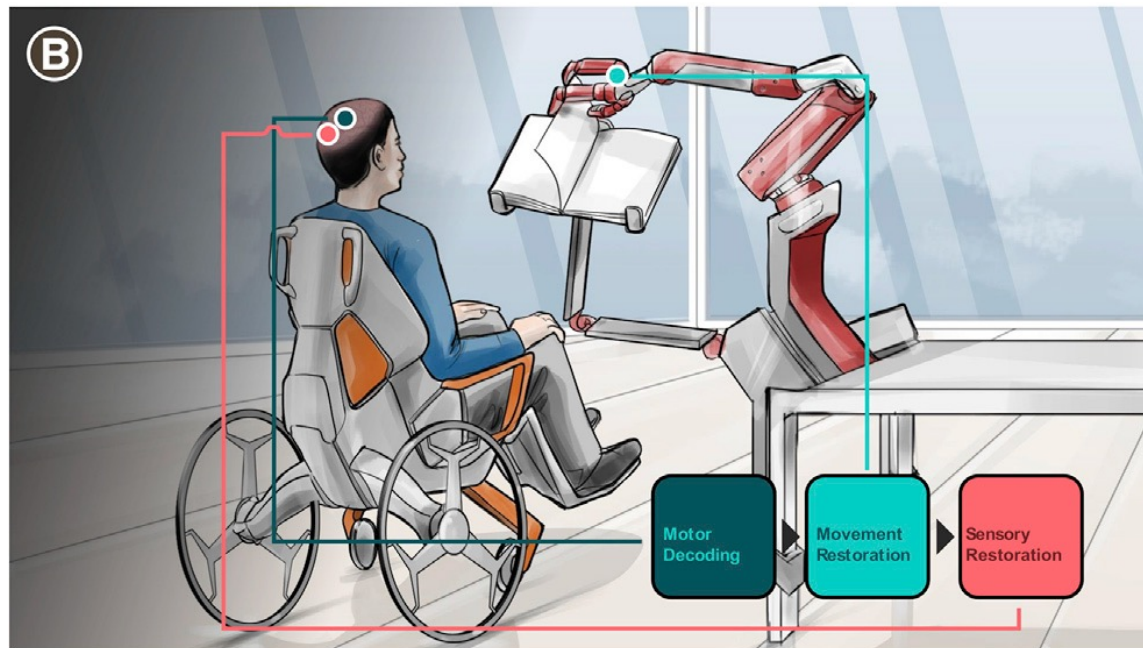
Unified framework for neuroprostheses designed to restore **sensory-motor functions of the upper limbs**, structured around three core components:

- **Motor-decoding** block
- **Movement-restoration** block
- **Sensory-feedback-restoration** block.

Together, these components aim to replicate the natural human sensory-motor control loop. Each block incorporates established neurotechnological modules tailored to restore specific functions for various disabilities.



Bidirectional bionic hands to restore sensory and motor functions in patients with hand amputations



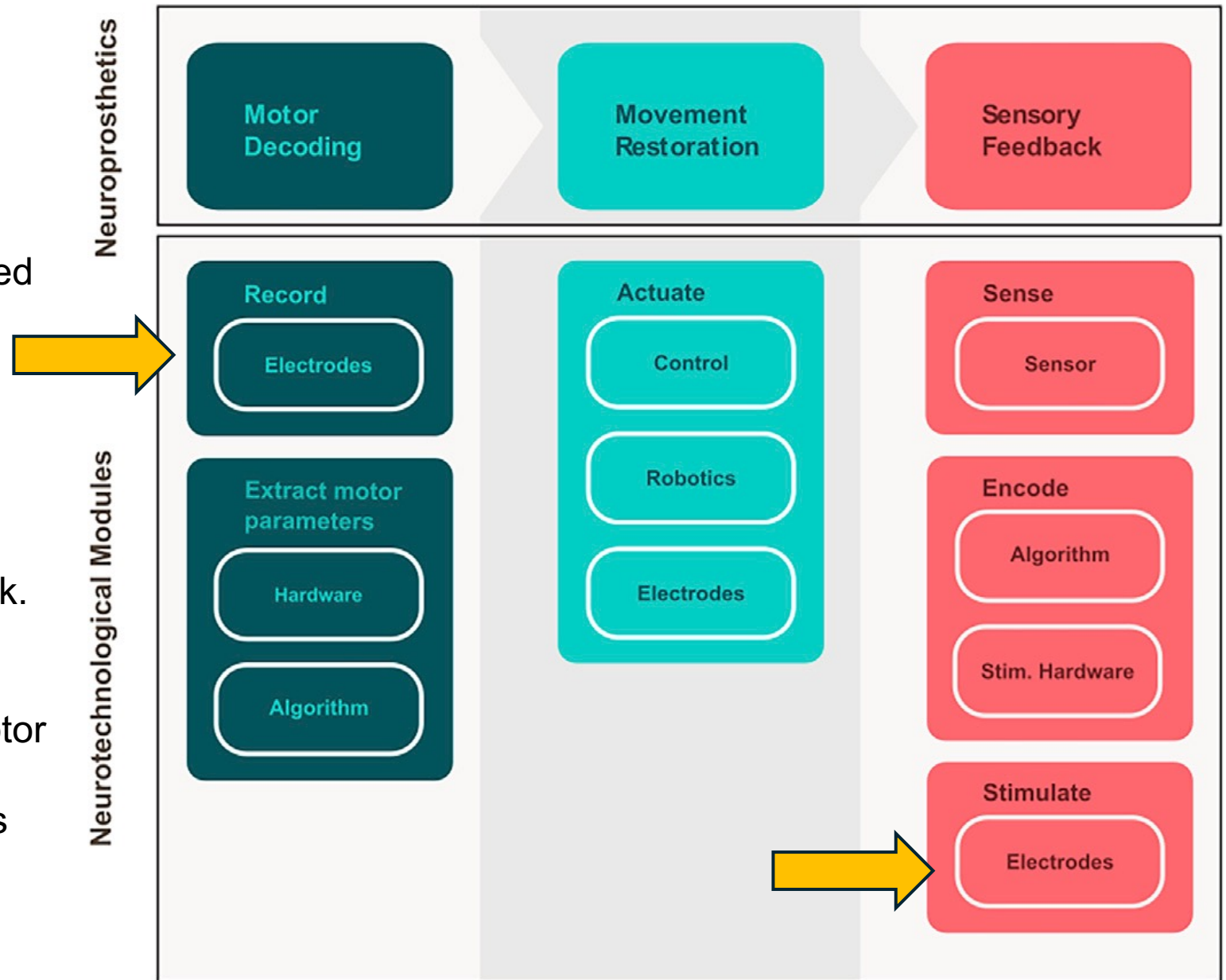
Bidirectional brain machine interfaces (BMIs) in patients with traumatic lesions or neurological disorders of the spine or above

Neuroprostheses

Unified framework for neuroprostheses designed to restore **sensory-motor functions of the upper limbs**, structured around three core components:

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- **Movement-restoration** block
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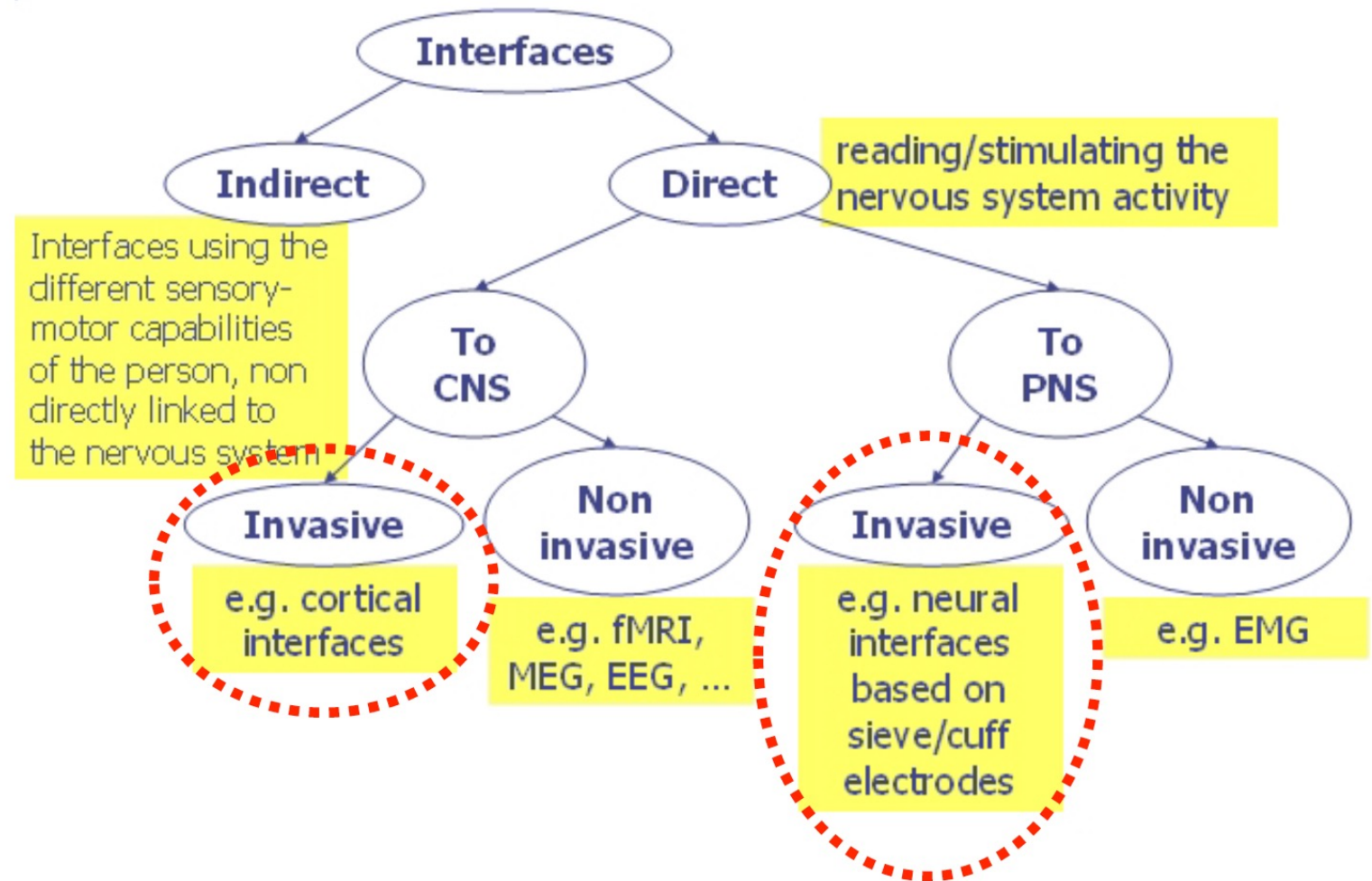


Neural interfaces

(Bio)artificial devices able to (re)create a bidirectional natural link with the (central and peripheral) nervous system

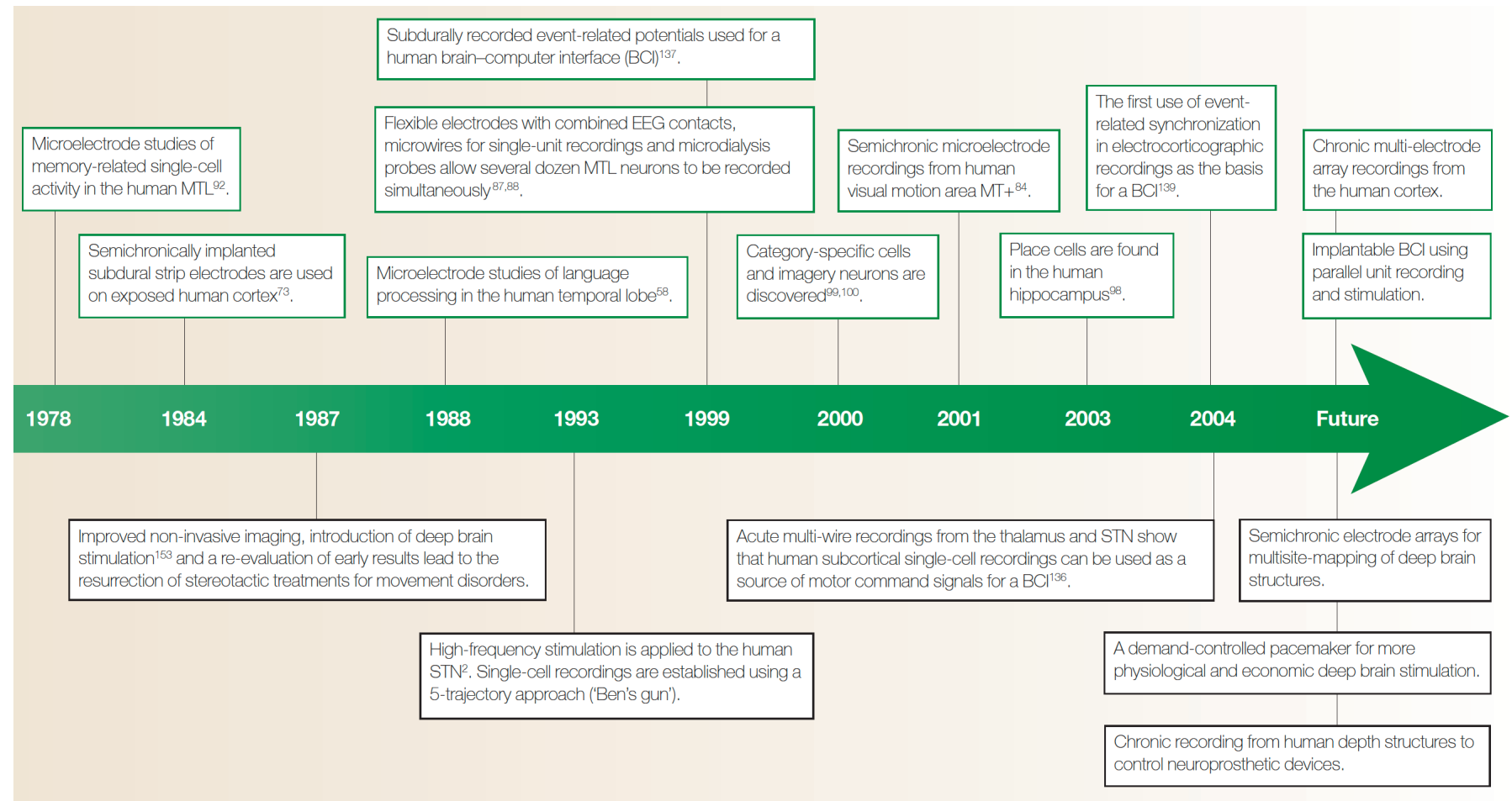
Nerve fibres are not “wires” or “cables” in the engineering point of view, but long, sensitive structures consisting of biological membranes with multiple receptors and delicate interactive elements for on-line sensing the environment and transmitting information via molecules, potentials and changes in the chemo-electrical activity

(Heiduschka and Thanos, Progress in Neurobiology, 1998)



Neural interfaces

Although non-invasive methods such as functional magnetic resonance imaging, electroencephalograms and magnetoencephalograms provide most of the current data about the human brain, their resolution is insufficient to show physiological processes at the cellular level.



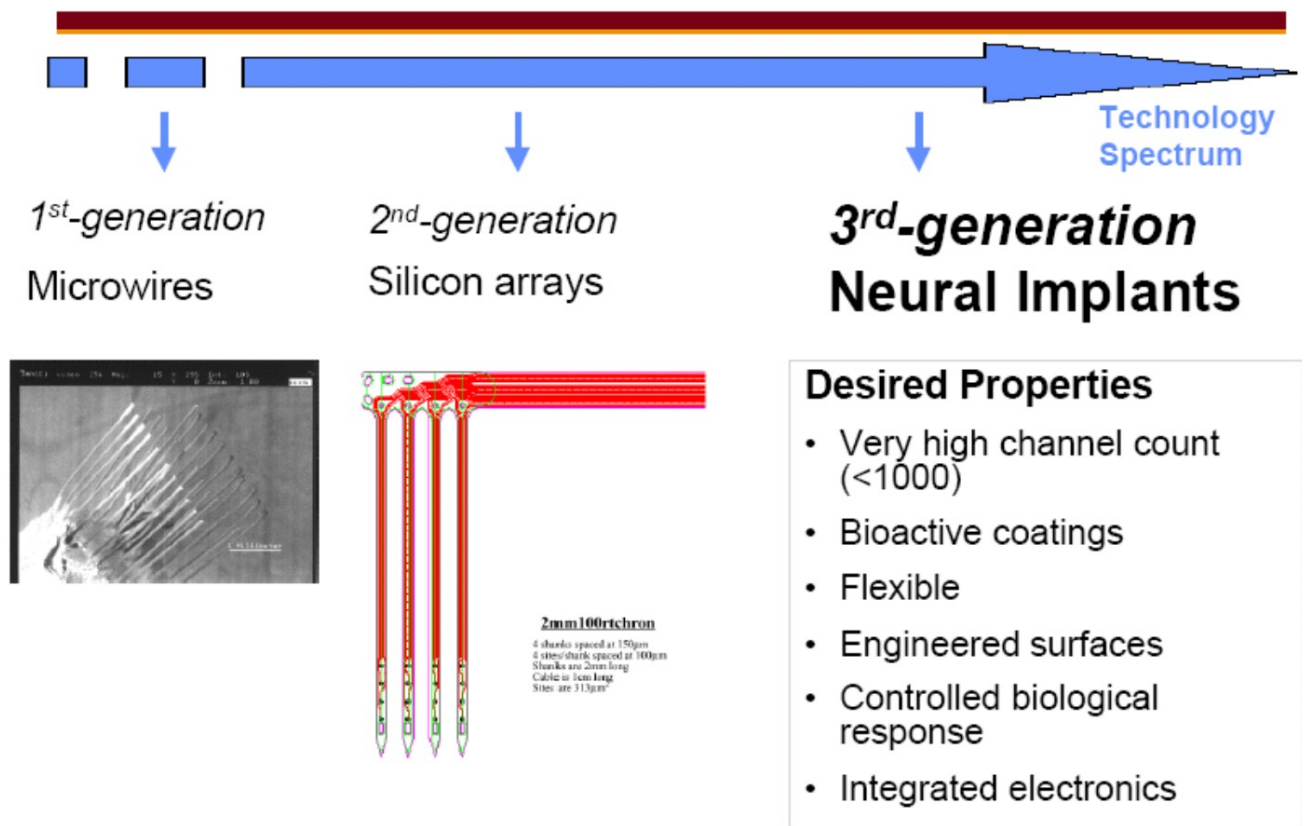
Interfaces with the CNS

Requirements for (cortical) invasive interfaces

Requirement	Primary design considerations
Safety	
Minimally invasive	Minimize size and access route
Safe to implant	Base on known surgical techniques
Fully implantable	Integrated electronics with wireless communication
Biocompatible	Use appropriate materials; Use appropriate interfaces
Efficacy	
Real-time cortical signals having 'sufficient' information content per application needs	Reliable & stable spike recordings from neural ensembles.
Integration with neuroprostheses & brain-computer interface devices	Defined signal interface between implantable probe system and external instrumentation
Stable and long-lasting cortical signals	Optimized probe design; Incorporation of intervention strategies for device-tissue interfaces
General	
Extensible	Modular design to provide progressive development path
Cost effective	Leverage existing devices, materials, and techniques.

Kipke, IEEE EMBS, 2004

Evolution of (cortical) invasive interfaces

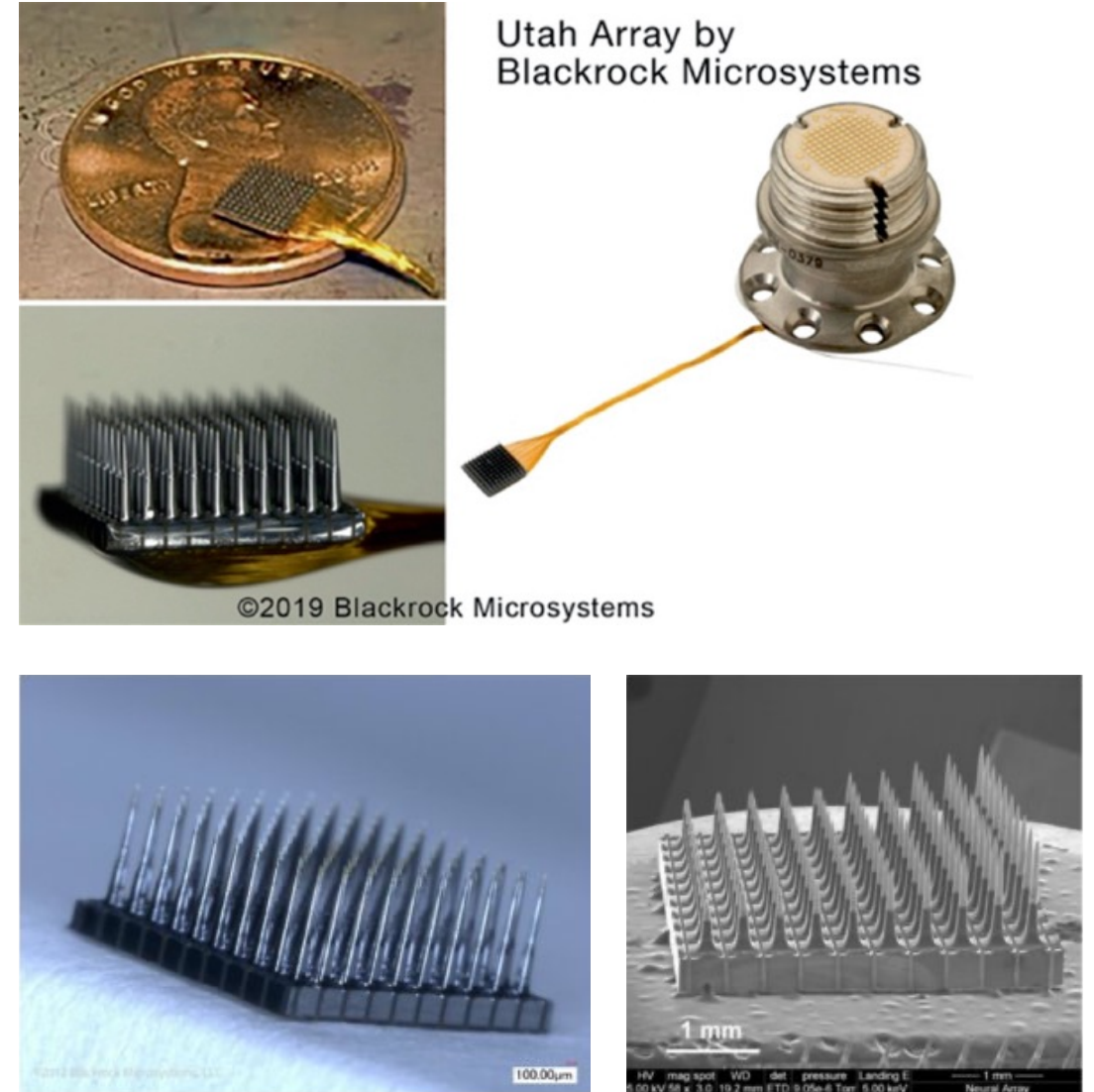


Interfaces with the CNS

- Neural sensors require integration with an electronic interface, forming part of a broader system that includes signal processors, wireless radios, and decoders, either on or off the body.
- Engineers must focus on optimizing **power** (measured in watts), **performance** (like precision in motor control tasks), and **packaging** (size and location), with all designs constrained by strict safety standards.
- For example, a 100-channel wireless neural recording device must adhere to FDA regulations limiting temperature increases and FCC (US Federal Communications Commission) rules on radio-frequency power, ensuring minimal heating and safe operation inside the body.

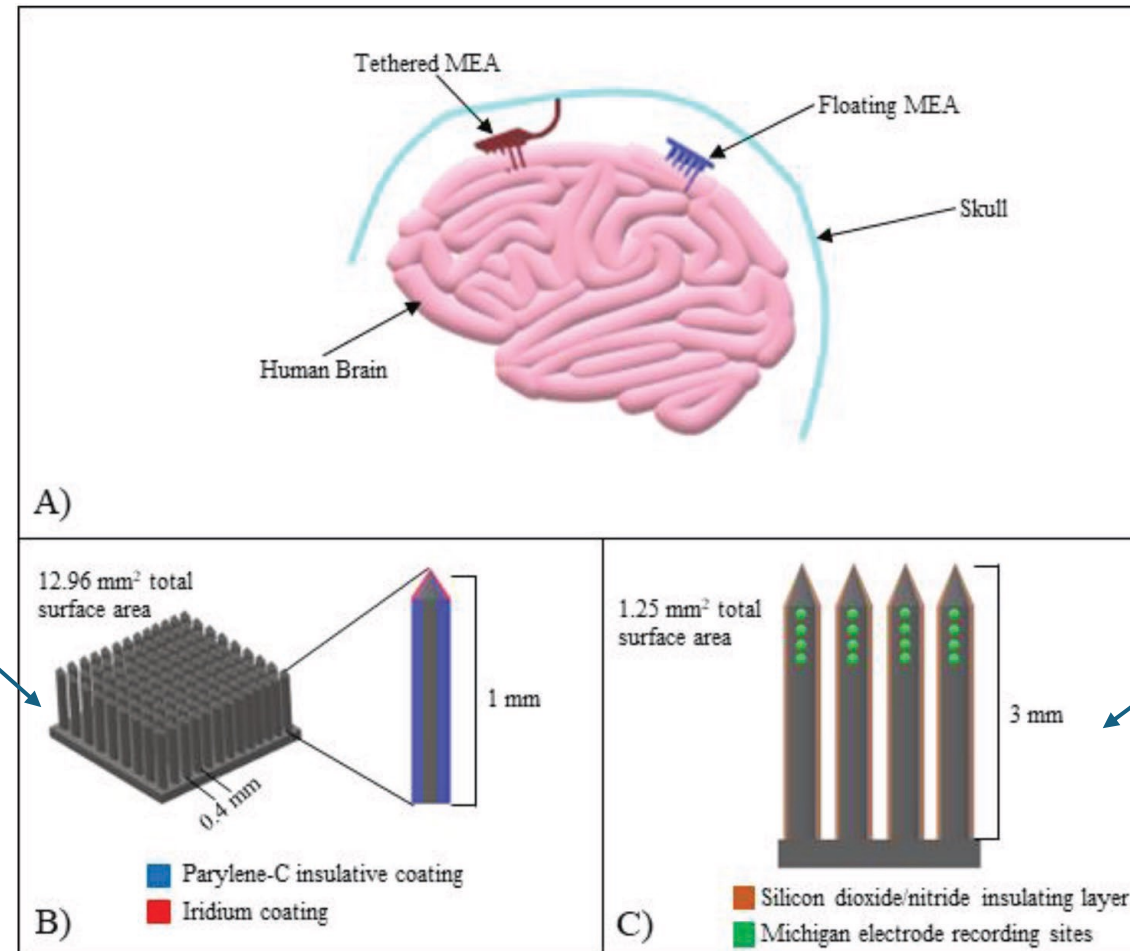
Interfaces with the CNS

- The introduction of silicon-based multichannel microelectrode arrays (MEAs) in the 1980s and 1990s, like the **Utah** and **Michigan** arrays, revolutionized the study of neural dynamics in the cortex.
- Unlike traditional single-wire electrodes, MEAs can record from many neurons simultaneously, which has led to deeper insights into how groups of neurons work together.
- This advancement, combined with improved computing, enabled models that predict neural population behavior and laid the groundwork for the first functional, brain-controlled prosthetics in humans.



Interfaces with the CNS

The MEA configurations include different designs for electrode arrays used in neural interfaces. Distinction between tethered and floating MEAs: tethered MEAs are fixed to the skull after implantation, limiting movement, while floating MEAs can move with the brain.



The Utah MEA configuration features grids of silicon electrodes, each 1 mm long with 0.4 mm spacing and a surface area of 12.96 mm², coated with Parylene-C and iridium.

The Michigan MEA configuration consists of a quad-shank arrangement with 16 recording sites, each shank measuring 3 mm in length and coated with silicon dioxide/nitride, offering a total surface area of 1.25 mm².

Interfaces with the CNS

Scaling up MEAs for more extensive neural interfacing presents new challenges:

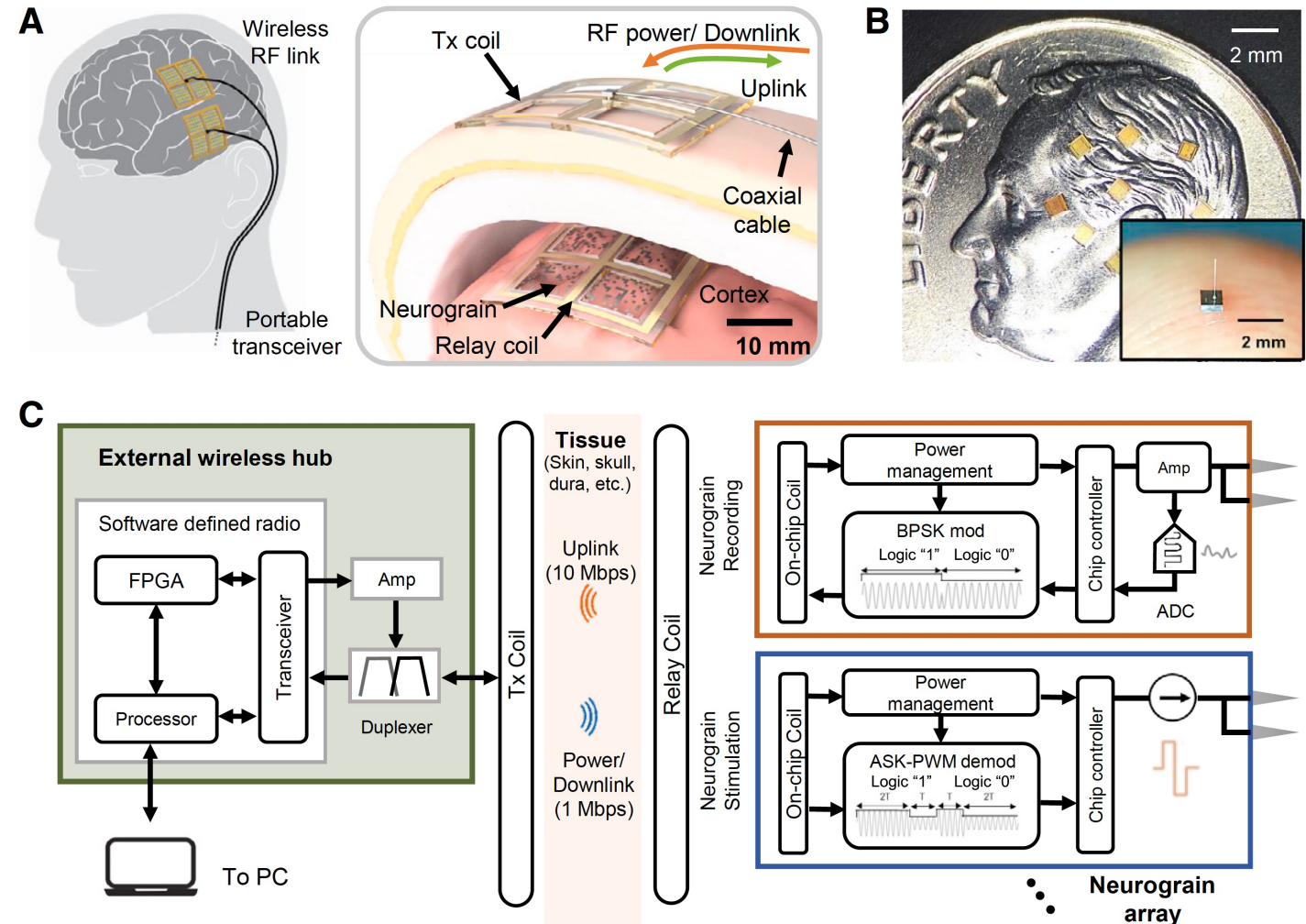
- While the current design of MEAs can displace a small amount of brain tissue, a larger array, such as one with 3,000 electrodes, would take up much more space, potentially causing damage.
- There are also technical challenges in connecting these arrays to external devices, as current wire connections can be problematic during surgery.
- Finally, while MEAs can last for years, degradation within the body and biocompatibility concerns limit their longevity.
- Recent research in 3D nanostructured materials and improved fabrication methods shows promise.
- Using MEAs across multiple brain regions has already produced valuable data, enhancing our understanding of cortical circuits and improving neural decoding capabilities for motor tasks, sensory input processing, and even speech decoding in both humans and animals.

Interfaces with the CNS

A wireless microsensor system operates as a spatially distributed network, utilizing an external transcutaneous RF power and data link to connect an implanted ensemble of microchips.

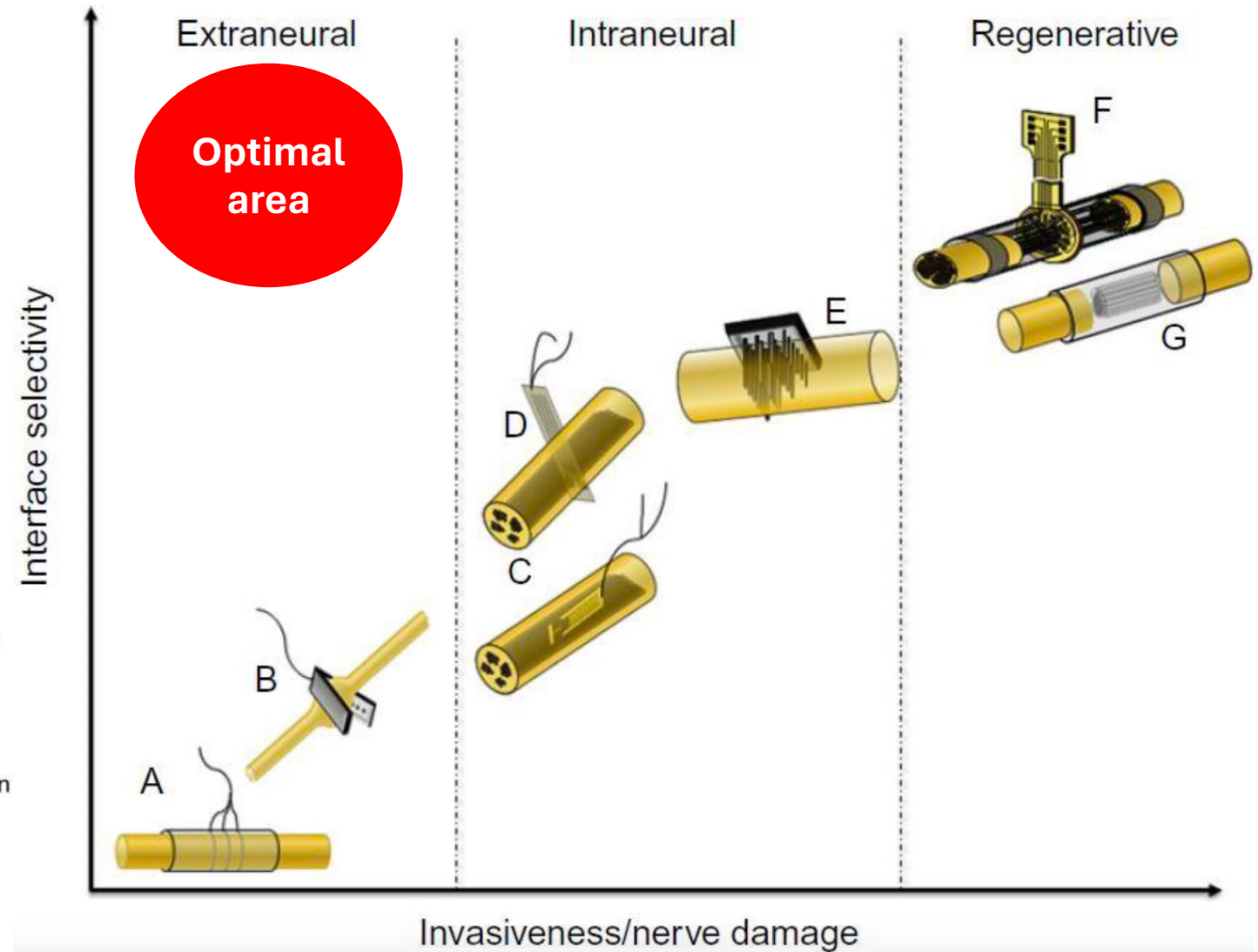
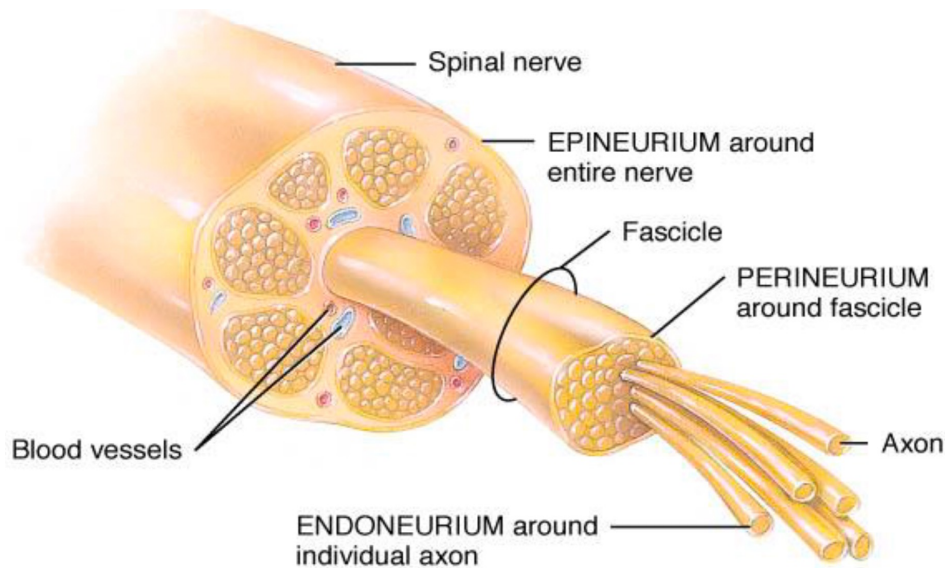
These microchips, each measuring $650\ \mu\text{m} \times 650\ \mu\text{m} \times 250\ \mu\text{m}$, are small enough to fit on a US quarter, with an optional post-processed integrated microwire for deeper intracortical access.

The overall electronic system includes an external RF hub, an inductive coupling link, and key features of the neurograins for both recording and stimulation.

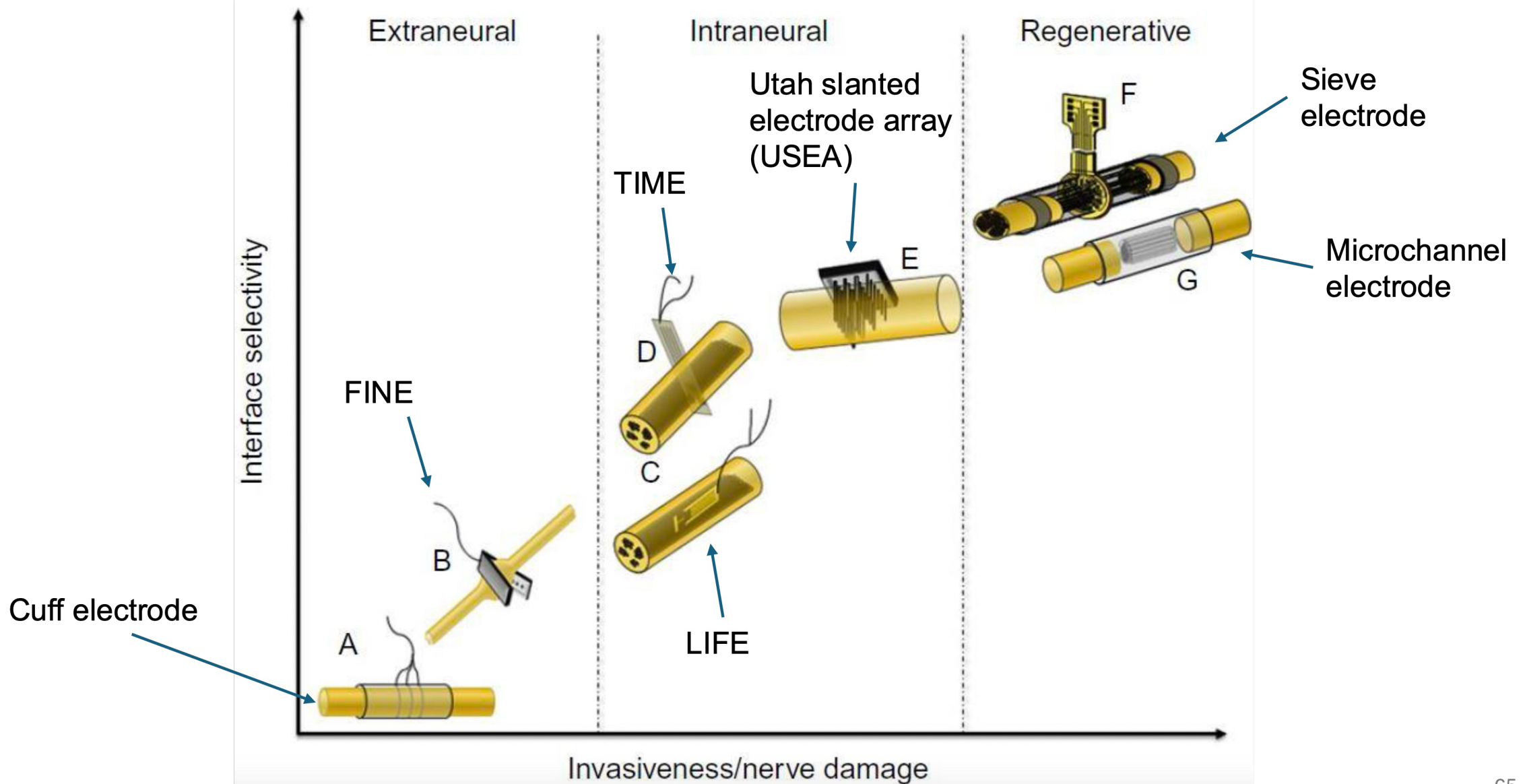


Interfaces with the PNS

The PNS has a very hierarchical structure and the connection we want to re-establish could be difficult to achieve since we want to interface very deep structures



Interfaces with the PNS



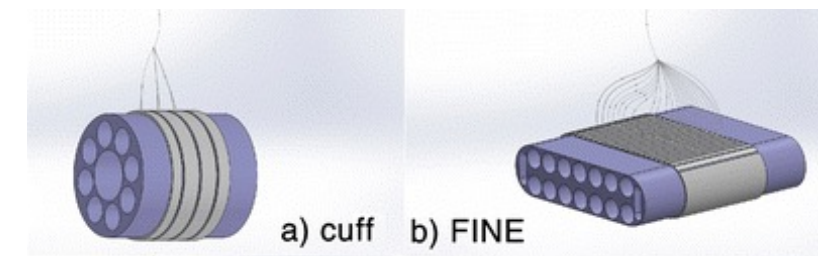
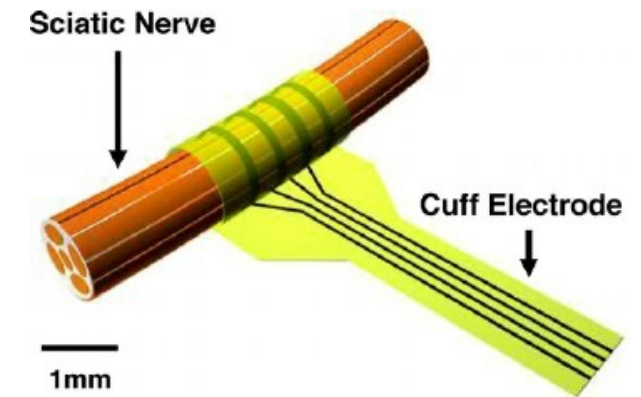
Extraneural electrodes

Extraneural electrodes are non-invasive devices that do not penetrate the nerve, minimizing the risk of nerve damage compared to intraneural electrodes.

Their distance from the nerve fibers reduces the specificity of the information they collect.

Cuff electrode: wraps around the nerve with electrical contacts on the inner surface, isolating the electrode from signals like electromyograms (EMGs). While nerve cuff electrodes **are less invasive and easier to implant**, their recording **selectivity is limited**, prompting efforts to improve this feature.

Flat Interface Nerve Electrode (FINE): reshapes the nerve by applying small forces, bringing nerve fibers closer to the electrode surface, **improving recording selectivity** without causing damage. FINEs have been used successfully for stimulation in humans and selective recordings in animals, with the composite FINE offering similar benefits with added flexibility.



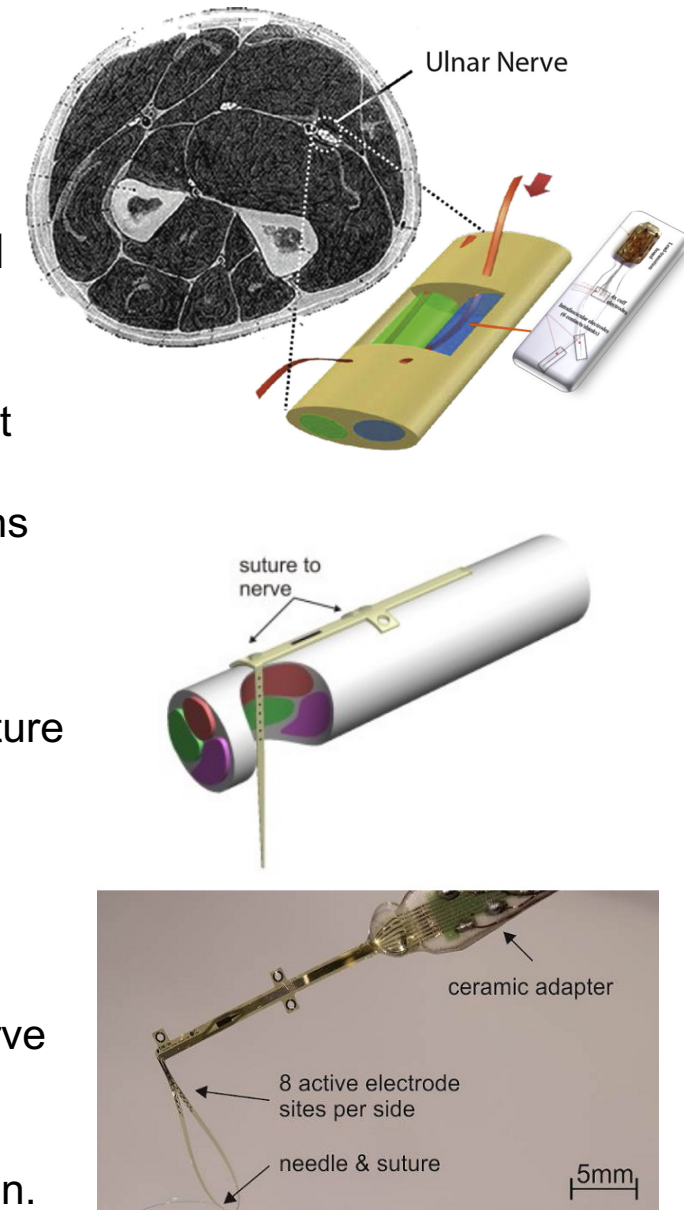
Intraneural electrodes

Intraneural electrodes penetrate the nerve, offering higher spatial selectivity than extraneural electrodes, but they also come with a higher risk of nerve damage and are more challenging to implant.

LIFE (Longitudinal Intrafascicular Electrode): consists of thin, flexible, insulated wires that are inserted longitudinally into the nerve to record from small regions. While LIFE electrodes offer excellent functionality for localized areas, they become impractical when multiple regions need to be recorded.

TIME (Transverse Intrafascicular Multichannel Electrode): inserted transversely into the nerve and can record from multiple sites across a broader area, enhancing the ability to capture data from various nerve fibers or fascicles, similar to implanting several LIFE electrodes.

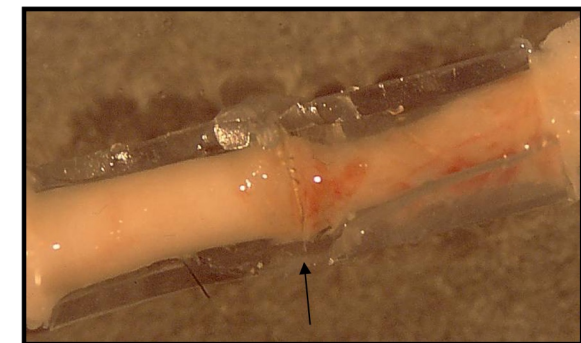
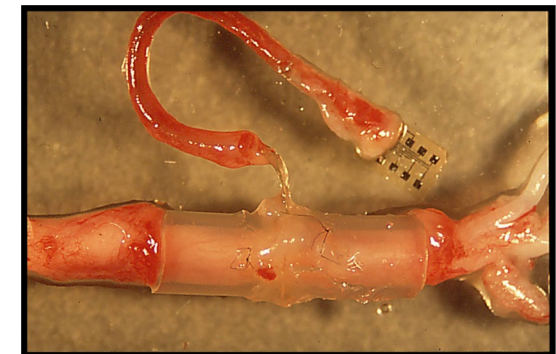
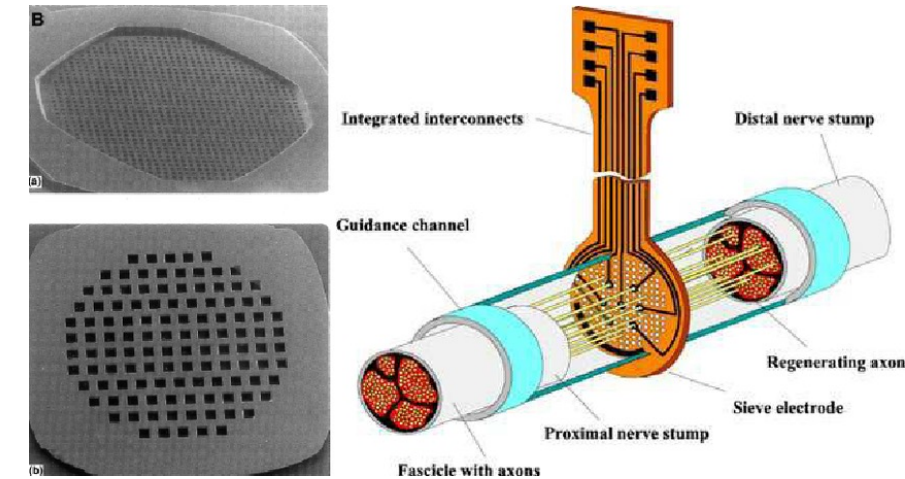
MEAs (Microelectrode Arrays), with their numerous recording locations, are ideal for monitoring large areas within a peripheral nerve, though their rigid structure and potential tethering forces make them highly invasive and prone to damaging the nerve. The slowly penetrating intrafascicular nerve electrode (SPINE) offers a middle ground, resembling a nerve cuff but with internal plates that gently separate the nerve's layers for better selectivity while being less invasive than other intrafascicular designs. Despite the advantages of intraneural electrodes for selective recordings, challenges with long-term stability and chronic use remain.



Regenerative electrodes

Regenerative electrodes are a type of intraneural electrode designed to interface with severed nerve stumps, promoting nerve fiber regeneration through an array of holes with electrical contacts.

- These electrodes enable selective recording and stimulation of individual or small groups of nerve fibers. Their use in humans is limited due to the significant nerve damage required and the time needed for nerve fibers to regenerate.
- Polyimide sieve electrodes implanted in the rat sciatic nerve allowed axon regeneration through the array of holes, with positive results in all animals. As implantation time increased from 2 to 6 months, there was a progressive increase in the number and maturation of regenerated axons.
- However, at 6 to 12 months, some rats experienced compressive axonopathy, leading to a loss of regenerated fibers and a decline in target reinnervation.
- The maintenance of regenerated axons is challenging, particularly in the absence of distal target organs, such as in amputated limbs.



Nerve regenerated through the regenerative electrode